

Unit 12 Overview: Segment Relationships in Circles



Unit Narrative

In this Unit, students will get a deeper understanding of circles. Students will solve mathematical and real-world problems involving the length of secants, tangents, and chords of a given circle. Students will need to be able to recall the Pythagorean Theorem to help build connections on new material, such as tangent segments of a circle. Students will continue to build on their understanding of circles by deriving the equation of a circle and being able to graph and identify key features.

Section 1 consists of Lessons 1 through 6, and will focus on the segments of a circle. Students will learn which theorems to apply to the specific segments of a circle, and how to solve for the lengths of given segments. Students will strengthen their understanding of secants, tangents, and chords while developing new relationships.

Section 2 will focus on the equation of a circle. Students will derive the equation of a circle in center-radius form, and learn how to convert between center-radius form and general form. Students will build on their knowledge of completing the square from Algebra 1 in order to be able to convert from general form to center-radius form of a circle. The unit ends with students being able to graph and identify key features of circles in general and center-radius form.

Consider what differentiation might look or sound like for this unit, specifically taking into consideration some students may have worked with the Pythagorean Theorem and some have not. Students will also need to be comfortable with solving for variables, as it is critical when finding the length of line segments.

Unit At-A-Glance

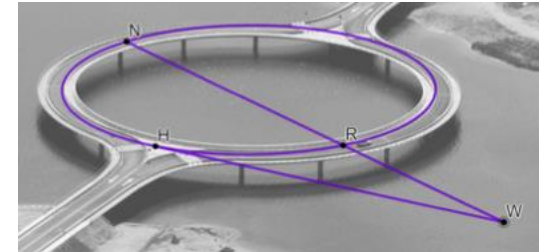
Section 1: Line Segments of Circles	Benchmarks Addressed	Lesson 1	Solving Problems With Lengths of Intersecting Chords
	MA.912.GR.6.1	Lesson 2	Solving Problems With Lengths of Two Secants
		Lesson 3	Solving Problems With Secant and Tangent Segments
		Lesson 4	Solving Problems With Lengths of Two Tangents
		Lesson 5	Pythagorean Theorem and Tangent Lines
		Lesson 6	Lengths of Secants, Tangents, Segments, and Chords in Circles
Section 2: Equations, Graphs, and Key Features of a Circle	Benchmarks Addressed	Lesson 7	Deriving the Equation of a Circle
	MA.912.GR.7.2	Lesson 8	General Form of a Circle
	MA.912.GR.7.3	Lesson 9	Graphing Circles



Differentiation

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Engagement <i>The “WHY” of learning</i>	- Real-world examples act as catalysts for engaging students in the “why” behind their learning. Consider what real-world applications can look like for you and your students, based on your local community context and your unique students and classroom. Some examples from include: - Finding distance on a softball or baseball field using the pitcher’s mound and basepaths - Finding distance between islands using various landmarks (or the Bermuda Triangle) - Finding the distance between certain locations at your school or in your community
Representation <i>The “WHAT” of learning</i>	- In Lessons 1-5 encourage students to make meaning of what they are processing using models and diagrams. Provide construction paper, scissors, markers, patty paper, and more for additional entry points as students learn. - In Lessons 7-9 Use Desmos, GeoGebra, and other digital platforms provide a plethora of digital manipulatives for students to explore and make sense of information in different ways.
Action & Expression <i>The “HOW” of learning</i>	In Lesson 6 consider using 12.6.1 Warm-Up as a launch for an extension project that will provide a variety of options and pathways for students to demonstrate their learning. Students can submit a project with no parameters to encourage creativity. The project must capture what they have learned about secants, tangents, and chords so far in the unit (Lessons 1-6). The only stipulation is that it needs to either be in a different language or based on a different country. The emphasis on a completely different context will ignite new pathways for students.



ELL Information

Know who your ELL students are and their proficiency in English. This can be obtained from the ESOL/bilingual teacher, guidance counselor, social worker, or administration. It is important to note that their proficiency in listening, reading, speaking, and writing may vary. As with all students, set high expectations. Focus on vocabulary and language development, guided interaction, authentic assessment, explicit instruction, meaning-based context and universal themes, modeling, graphic organizers, visuals, and group work.

- Modeling with colored pencils or markers is a helpful way to show students to see the connection between information in the question and the line segments.
- Create a new visual aid, chart paper sized or interactive notebook sized, with a large circle, illustrating several of the vocabulary words. Because students are already familiar with some of the vocabulary words, try a word bank activity and have students label the different parts, such as radius, diameter, center, chord, tangent, secant, etc.
- In word problems and on diagrams, have students highlight important vocabulary words, such as center, radius, diameter, tangent, perpendicular, etc.

Additional Differentiated Support

Scaffolding Resources	On-Ramp
• Play memory games or matching games with images of circles and line segments with the type of line segment shown. • Mark Up Text can be used to help students approach more complicated and wordy problems, as well as provide visual cues for additional feedback between students and teachers. • Color coding given information and the corresponding parts on the line segments can provide visuals for students. • Example Activity: Explore the relationships of line segments and circles by providing students with pipe cleaners. Students can draw a circle, place the pipe cleaners from a given description (chords, secants, tangents, secant and tangent), then measure the pipe cleaners to build a connection between the line segment and its length.	Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish. • Area and Circumference of a Circle • Pythagorean Theorem



Section Overview

Section 1: Line Segments of Circles (Lessons 1-6)

Benchmarks Addressed	Learning Targets	
MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle. <i>Clarification 1: Problems include relationships between two chords; two secants; a secant and a tangent; and the length of the tangent from a point to a circle.</i>	Lesson 1	I can solve mathematical and real-world problems involving lengths formed by the intersection of two chords.
	Lesson 2	I can solve mathematical and real-world problems that involve the lengths formed by the intersection of two secants.
	Lesson 3	I can solve mathematical and real-world problems that involve lengths formed by the intersection of a secant and a tangent segment.
	Lesson 4	I can solve mathematical and real-world problems that involve lengths formed by the intersection of two tangents.
	Lesson 5	I can use the Pythagorean Theorem to solve mathematical and real-world problems involving the length of a tangent segment to a circle.
	Lesson 6	I can solve mathematical and real-world problems involving the lengths of secants, tangents, segments, or chords of a circle.

Section 2: Equations, Graphs, and Key Features of a Circle (Lessons 7-9)

Benchmarks Addressed	Learning Targets	
MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features. <i>Clarification 1: Instruction includes using the Pythagorean Theorem and completing the square.</i> <i>Clarification 2: Within the Geometry course, key features are limited to radius, diameter and the center.</i>	Lesson 7	- I can use the Pythagorean Theorem to derive the equation of a circle. - I can determine key features of a circle given the graph or equation. - I can write the equation of a circle using key features.
	Lesson 8	- I can rewrite the equation of a circle from general form to center-radius form. - I can determine key features of a circle given the equation in general form. - I can write the domain and range using inequality notation, interval notation, or set-builder notation.

MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of the context.

- *Clarification 1: Key features are limited to domain, range, eccentricity, center and radius.*
- *Clarification 2: Instruction includes representing the domain and range with inequality notation, interval notation or set-builder notation.*
- *Clarification 3: Within the Geometry course, notations for domain and range are limited to inequality and set builder.*

Lesson 9

- I can graph a circle and determine its key features when given an equation of a circle.
- I can solve mathematical and real-world problems that are modeled with an equation of a circle.



Vertical Alignment

8th Grade Unit 5
Relationships in Triangles



Geometry Unit 12
Segment Relationships in Circles

Prior Course(s)	Course Level
MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.	MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle. MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features. MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of context.

12.1 Solving Problems With Lengths of Intersecting Chords

Lesson Introduction

 Benchmark Focus	MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.		 Learning Target	I can solve mathematical and real-world problems involving lengths formed by the intersection of two chords.
 Benchmark Coherence	Building On: N/A		Working Toward: N/A	
 Essential Questions	- What is the relationship between the lengths of two intersecting chords? - What is the relationship between two chords if the two chords are equidistant from the center of a circle?			
 Lesson Narrative	This lesson activates prior knowledge of perpendicular bisectors, chords, and triangle similarity theorems. Students will be introduced to the properties of chords in circles through exploration. The primary learning target in this lesson focuses on two chords being equidistant from the center of the circle and intersecting chords.			
 Component Summary	12.1.1 Warm-Up (~5 min.)	Notice and wonder about bowling ball pyramid (<i>SE p. 231</i>)	12.1.3 Bring It Together (10 min.)	Chord relationships in circles (<i>SE p. 234</i>)
	12.1.2 Exploration (15-20 min.)	Using patty paper to explore perpendicular bisectors on chords (<i>SE p. 232</i>)	12.1.4 Guided Practice (10 min.)	Practice finding lengths of chords (<i>SE p. 236</i>)
	12.1.5 Wrap-Up (~5 min.)	Reflection on mastery of learning targets (<i>SE p. 237</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Chord - Diameter - Radius - Congruent - Perpendicular - Bisect		- Patty Paper (Optional) Colored Pencils or Markers	
			Additional Preparation	
			12.1.2 Exploration has students copying three circles and constructing their perpendicular bisectors. Prepare patty paper for each student.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not recall the properties of a perpendicular bisector. - Students may add the intersecting chords together, rather than find the product of their segments. - Students may struggle acquiring new vocabulary/formulas and connecting them with known vocabulary/formulas.	- Provide colored markers or pencils to students to use in question 2 of 12.1.2 Exploration. They can highlight the different parts of the circle they are working with (chords, radii, perpendicular bisector, triangle). - Consider illustrating the three chord relationships on a separate piece of chart paper that can be hung in the classroom as a reference for the students.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 12.1.2 Exploration provides an opportunity for students to use this routine. However, the questions are designed to spark thinking for the individual learners. Allow each student time to make a conjecture about the answer to the questions, share their reasoning with a partner, and have a variety of students share with the whole class.	MA.K12.MTR.1.1: Participate in Effortful Learning The expectation of 12.1.2 Exploration is to develop understanding through the use of manipulatives, visual models, discussion, estimation, and drawings. Continue to encourage students to use correct terminology when discussing with partners to help deepen their understanding.
 Differentiate Instruction	12.1.2 Exploration provides a variety of entry-points for learners to engage with the content. Part a provides a hands-on component along with an individual conjecture opportunity. Question 1b offers space for verbal reasoning with a partner, and question 1c positions students as mathematicians, expressing what they know from question 1a and 1b into a conjecture. Some students may prefer question 1a, and independently process information, while other students might prefer question 1b, and verbally process in order to make sense of the information. Ensure equitable time for all three parts during the component.	
Lesson Notes:		

12.1.1 Warm-Up: Notice and Wonder

Component Overview: Students reflect on and then share what they notice and wonder about a photograph of a pyramid of bowling balls. The photograph is intended to inspire student curiosity in the upcoming lesson. Ask students how the photograph may connect to exponents.



12.1.1 Warm-Up: Notice and Wonder

1. A picture of a bowling-ball pyramid is shown.



Notice and Wonder

Prompt students to reflect on what they notice and wonder about the photograph. Call on a variety of students to share what they noticed and wondered. This activity provides a low floor, as all students can participate, and a high ceiling, as students can be challenged to think of multiple things they may notice and wonder.

- a. What do you wonder?

Answers will vary. How many bowling balls are there? How was the pyramid made?

- b. What do you notice?

Answers will vary. The bowling balls are different colors. Each row has the same number of bowling balls.

- c. Write a question about the pyramid that a 5th grader could answer.

How many bowling balls are in the pyramid?

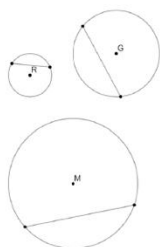
12.1.2 Exploration: Chords

Component Overview: Students will use patty paper to copy circles and then construct a perpendicular bisector on the chord inside each given circle. Students will write a conjecture and share with a partner. They will then use prior knowledge of perpendicular bisectors and similar triangle properties to explore the effects they have on the chords of a circle.



12.1.2 Exploration: Chords

1. Circles R , G , and M are shown. Each circle has a chord. Use patty paper to copy the circles. Use the patty paper copy to construct a perpendicular bisector for each chord by folding.



- a. Write a conjecture about the perpendicular bisector of a chord of a circle.

Answers will vary. The perpendicular bisector of a chord in a circle passes through the center of the circle.

- b. Share your conjecture with your partner. Listen to your partner's conjecture and write a summary. Have your partner initial that it is summarized correctly.

Summary of Partner's Conjecture	Partner's Initials
<i>Answers will vary.</i>	

- c. Review your conjecture and make any revisions, if needed.

Answers will vary.



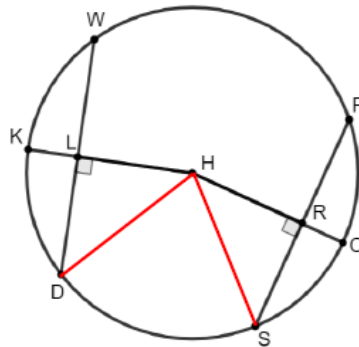
What is the difference between a chord, a radius, and a diameter? Can a chord be a radius? Why or why not? Can a diameter be a chord? Why or why not?



Teachers may need to review the construction of a perpendicular bisector. Students are not expected to have a perfect mathematical knowledge of chords and their properties at this point. This concept is being developed through exploration, so encourage informal conjectures.

12.1.2 Exploration: Chords (continued)

2. Circle H is shown, where $\overline{KH}$ and $\overline{CH}$ are radii of the circle. $\overline{WD}$ and $\overline{PS}$ are chords of the circle equidistant from the center H . Therefore, $\overline{LH} \cong \overline{RH}$.



- Using your conjecture about perpendicular bisectors, complete each congruence statement.
 $\overline{DL} \cong \overline{WL}$
 $\overline{PR} \cong \overline{SR}$
- Draw a line segment from point D to point H and a line segment from point S to point H .
See above.
- Justify why $\overline{DH}$ and $\overline{SH}$ are congruent.
They are congruent because they are both radii of the circle.
- Given that $\overline{LH} \cong \overline{RH}$ and $\overline{SH} \cong \overline{DH}$, what triangle congruence theorem justifies that $\triangle DLH$ and $\triangle SRH$ are congruent?
Hypotenuse Leg Theorem
- Since $\triangle DLH \cong \triangle SRH$, what is true about $\overline{DL}$ and $\overline{SR}$?
They are congruent and therefore have equal measures.
- What is true about chords $\overline{DW}$ and $\overline{SP}$?
They are congruent and therefore have equal measures.



Notice and Wonder

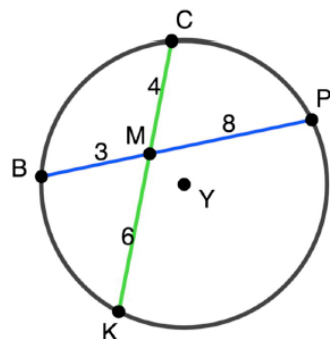
Consider providing students a few minutes to write down things they notice and wonder about the image before answering the questions. This will help students get familiar with the circle and chords prior to answering the questions.



Create a new visual aid, chart paper sized or interactive notebook sized, with a large circle, illustrating several of the vocabulary words. Because students are already familiar with some of the vocabulary words, try a word bank activity and have students label the different parts, such as the radius, diameter, center, and chord.

12.1.2 Exploration: Chords (continued)

3. Circle Y is shown, where points C , P , K , and B are on circle Y . Chords PB and KC intersect at point M .



Observations of the Blue Chord Measures	Observations of the Green Chord Measures
$3 \times 8 = 24$	$6 \times 4 = 24$
Commonalities in the Blue and Green Chords <i>Answers will vary. The products of the two pieces of the chords both equal 24.</i>	
What do you wonder about the relationship between the blue chord and the green chord? <i>Answers will vary. Will all intersecting chords of a circle cut the chords into two pieces that when multiplied will equal the same product?</i>	



For struggling students, consider providing a chart for students to look at to see the different possible operations for each chord measure.

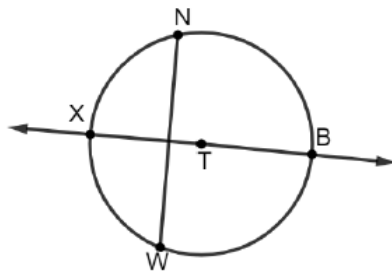
12.1.3 Bring It Together: Chord Relationships

Component Overview: Students will work individually, utilizing 12.1.2 Exploration as a guide. They will be introduced to the properties of the diameter of a circle that is perpendicular to a chord, and chords equidistant from the center. This component serves as a connection to their findings from 12.1.2 Exploration. Encourage students to use the correct vocabulary when discussing these new properties.



12.1.3 Bring It Together: Chord Relationships

- Circle T is shown with chord $\overline{NW}$ and secant $\overline{XB}$, where $\overline{XB}$ bisects $\overline{NW}$ and $\overline{XB}$ is a diameter of the circle.



What information is known about circle T , chord $\overline{NW}$, and the angles formed by $\overline{XB}$ and $\overline{NW}$?

Our previous conjecture stated that if a diameter bisects a chord, then the diameter is perpendicular to the chord so $\overline{XB} \perp \overline{NW}$.



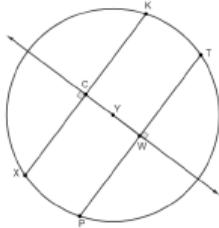
If a diameter of a circle is perpendicular to a chord, then the diameter bisects the chord and its arc.



Consider using students' conjectures from 12.1.2 Exploration to guide student understanding of the relationships between chords and perpendicular bisectors.

12.1.3 Bring It Together: Chord Relationships (continued)

2. Circle Y is shown with chords $\overline{XK}$ and $\overline{PT}$, where $\overline{CY} \cong \overline{WY}$.



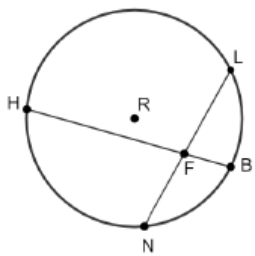
What information is known about circle Y , chords $\overline{XK}$ and $\overline{PT}$, and the angles formed by $\overline{CW}$?

Since $\overline{CY} \cong \overline{WY}$, then $CY = WY$, therefore showing the two chords are equidistant from the center of circle Y . Since distance is the perpendicular distance then the two chords are congruent.



If two chords are equidistant from the center of a circle, then the chords are congruent.

3. Circle R is shown with chords $\overline{HB}$ and $\overline{LN}$ that intersect at point F .



What information is known about circle R , chords $\overline{HB}$ and $\overline{LN}$, and the angles formed?

$$(HF)(FB) = (LF)(FN)$$

$$m\angle HFN = m\angle LFB \text{ and } m\angle HFL = m\angle NFB$$



When two chords intersect each other inside a circle, the products of their segments are equal.



Allow students to engage in meaningful dialogue related to the properties of chords in questions 1-3 to ensure they understand each relationship and the difference between the properties.



Prompt students to make sense of this information in their own way. Put students in pairs to construct an example that illustrates the meaning of the properties in each blue box.

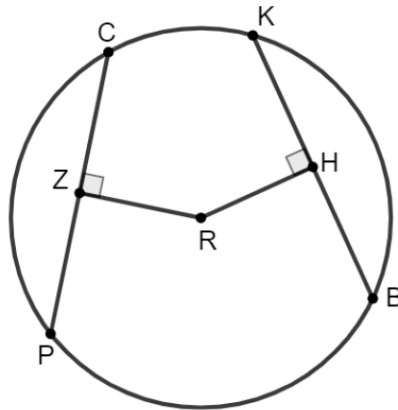
12.1.4 Guided Practice: Solving Problems Involving Intersecting Chords

Component Overview: Students will apply their conceptual understanding to solve questions involving intersecting chords. Students can complete this component independently and peer review their partners' responses.



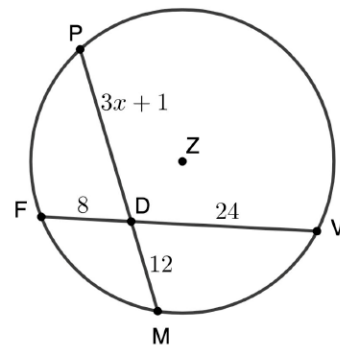
12.1.4 Guided Practice: Solving Problems Involving Intersecting Chords

- Circle R with points C , K , B , and P on the circle are shown with chords CP and KB equidistant from the center. Find the value of x , if $CP = (6x - 7)$, $KB = 35$, $ZR = HR$, and $m\angle CZR = m\angle KHR = 90^\circ$.



$$6x - 7 = 35, \quad x = 7$$

- In circle Z , chords PM and FV intersect at point D . Points P , F , M , and V lie on circle Z . Given $FD = 8$, $DM = 12$, $VD = 24$, and $DP = 3x + 1$, solve for x .
 $(3x + 1)(12) = 8(24)$
 $x = 5$



Use informal observations of student responses from 12.1.2 Exploration to inform how students will engage with this component.

- Students can collaborate on question 1 in pairs, using a Turn and Talk routine to facilitate discussion. In pairs, students then choose either question 2 or question 3 to complete before comparing responses.
- The teacher facilitates a student-centered discussion on question 1 before releasing students to attempt questions 2-3 on their own.
- The teacher pulls a small group for accelerated support, while the rest of the class independently attempts questions 1-3.



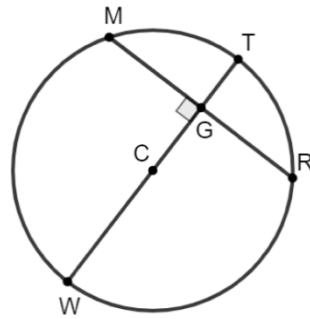
Describe the steps for solving a problem involving the lengths of intersecting chords within a circle. Create/produce an example for use in your explanation.

12.1.4 Guided Practice: Solving Problems Involving Intersecting Chords (continued)

3. Circle C is shown, where points T , R , W , and M are on the circle. The diameter, WT , is a perpendicular bisector of chord MR . If $MG = (y + 4)$ and $GR = (2y + 1)$, what is the length of the chord MR ?

$$y + 4 = 2y + 1 \quad y = 3$$

$$MR = 14$$



Will the intersecting chords theorem work if one of the chords is a diameter? Why or why not? Explain.



Consider offering students the option to color code each chord and their lengths given. This will help students identify which measures go with each chord.



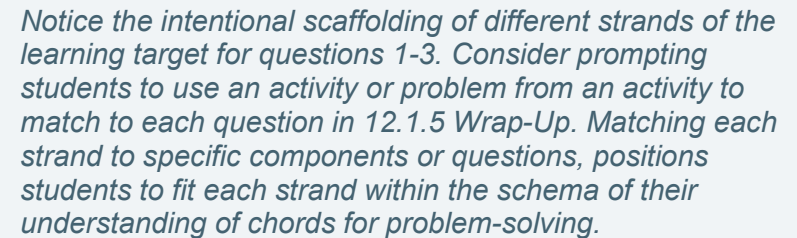
12.1.5 Wrap-Up: Reflection

Answers will vary.

- | | | |
|----------------------|------------------------|-----------------------|
| I need help. | I'm getting there, but | I understand this |
| I can't get started. | I need more practice. | and I feel confident. |

- | | | |
|----------------------|------------------------|-----------------------|
| I need help. | I'm getting there, but | I understand this |
| I can't get started. | I need more practice. | and I feel confident. |

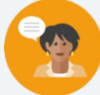
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|----------------------|------------------------|-----------------------|
| I need help. | I'm getting there, but | I understand this |
| I can't get started. | I need more practice. | and I feel confident. |



12.2 Solving Problems With Lengths of Two Secants

Lesson Introduction

 Benchmark	MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.		 Learning Target	I can solve mathematical and real-world problems that involve the lengths formed by the intersection of two secants.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is the relationship of the lengths formed by the intersection of two secants? - How can you find the lengths of segments formed by the intersection of two secants?			
 Lesson Narrative	This lesson will introduce the student to a secant line and its properties. Students will start with an exploration of given segment lengths and will complete a table about the lengths formed by the intersection of two secants. Students will work with a partner to make a conjecture and then write a formula about the lengths formed by the intersection of two secants. Students will be guided through a series of problems, including real-world examples. Students will conclude by trying problems on their own.			
 Component Summary	12.2.1 Warm-Up (~5 min.)	Would You Rather? with gold cubes (<i>SE p. 238</i>)	12.2.3 Guided Practice (10-15 min.)	Solving problems involving intersecting secant lengths (<i>SE p. 239</i>)
	12.2.2 Exploration (10-15 min.)	Explore intersecting secant lengths theorem (<i>SE p. 238</i>)	12.2.4 Your Turn! (5-10 min.)	Solving problems involving intersecting secant lengths independently (<i>SE p. 241</i>)
	12.2.5 Wrap-Up (~5 min.)	Error Analysis of student work finding lengths of intersecting secants (<i>SE p. 241</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> Secant <u>Additional Terms:</u> N/A		(Optional) Chart Paper (Optional) Markers or Colored Pencils	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	• Students may confuse a secant line and a tangent line. • Students may confuse multiplying and adding in the formula.	• Consider illustrating the theorem on a piece of chart paper that will be left hanging in the classroom as a reference for students. • If the students use an interactive notebook, consider having questions 1a-c from 12.2.2 Exploration available for them to paste into their notebook or have them copy questions 1a-c in their notebooks and color code the appropriate parts in the diagrams.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time 12.2.2 Exploration provides a structured and interactive opportunity for students to revise and refine both their ideas, and their verbal and written output. This routine provides a purpose for student conversation, as well as fortifies output. Throughout this process, students should be pressed for details, and encouraged to press each other for details.	MA.K12.MTR.4.1: Discussion and Reflection 12.2.2 Exploration allows the students to communicate mathematical ideas, vocabulary, and methods effectively. 12.2.2 Exploration will develop students' ability to justify methods and compare their responses to the responses of their peers.
 Differentiate Instruction	GeoGebra and other online tools provide digital manipulatives for engaging in content like chords, secants, and circles. 12.2.2 Exploration or 12.2.3 Guided Practice might be opportunities for students to engage with a digital manipulative to make connections between the lengths of secants and their relationship to each other.	
Lesson Notes:		

12.2.1 Warm-Up: Would You Rather?

Component Overview: Students are asked to determine which cube of gold they would rather have.



12.2.1 Warm-Up: Would You Rather?

1. Would you rather have ...

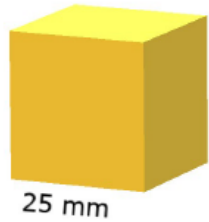
A cube of gold, 25 millimeters(mm) on each side ...

OR

Two cubes of gold, one is 24 mm per side, one is 1 mm per side ...



Provide the currency exchange rates for gold, silver, the US Dollar, cryptocurrency, and other country's currency for an engaging discussion on saving or investing money.



25 mm

$$25^3 = 15,625 \text{ mm}^3$$



24 mm

$$24^3 + 1^3 = 13,825 \text{ mm}^3$$



1 mm

2. Explain your choice.

Answers will vary. I would choose the 25mm cube of gold because it weighs more.

12.2.2 Exploration: Intersecting Secant Lengths

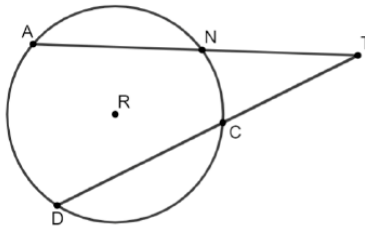
Component Overview: Students begin working with intersecting secant lengths. The outcome of this component is for students to be able to discover the formula used to find the lengths of intersecting secant lines. Consider giving students opportunities to analyze errors, engage in math talk, and collaborate.



12.2.2 Exploration: Intersecting Secant Lengths

A **secant** is a line that passes through two points on a circle. In the interior of the circle, a secant line forms a chord.

- Circle R is shown with points A , N , D , and C on the circle. Secants $\overline{AT}$ and $\overline{DT}$ intersect at point T outside the circle. $AN = 6$, $CD = 9$, $NT = 8$, and $CT = 7$.



- Complete the table.

Product	Work	Answer
$AN \cdot NT$	$AN \cdot NT = (6)(8)$	48
$AN \cdot AT$	$AN \cdot AT = (6)(14)$	84
$AT \cdot NT$	$AT \cdot NT = (14)(8)$	112
$DC \cdot CT$	$DC \cdot CT = (9)(7)$	63
$DC \cdot DT$	$DC \cdot DT = (9)(16)$	144
$DT \cdot CT$	$DT \cdot CT = (16)(7)$	112

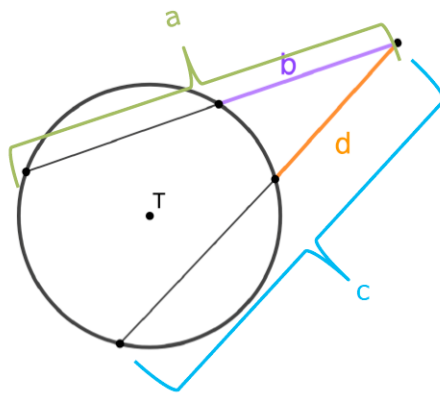


MA.K12.MTR.5.1: Patterns and Structure

Encourage students to use patterns and structure to help understand and build connection between the intersecting secant lines. This will support your students to develop generalizations based on similarities found among questions. This will also help students recognize patterns in the world around them and connect these patterns to mathematical concepts.

12.2.2 Exploration: Intersecting Secant Lengths (continued)

- b. Discuss the following with your partner.
- What do you notice about your answers?
 - What do you notice about the secant segments for the rows that have the same answer?
- c. With your partner, use the diagram of Circle T to write a formula that can be used to find the segment lengths of intersecting secants.



$$ab = cd$$



If two secants intersect outside a circle, then the product of the lengths of one secant segment and its external secant segment equals the product of the lengths of the other secant segment and its external secant segment.



Provide sentence frames to facilitate strong mathematical discussion:

- “When two secants intersect...”
- “The product of the lengths of one secant...”
- “When two segments of a secant are multiplied...”

After discussion, prompt students to write a summary of their findings.



Consider challenging students to write a formula if a and c were not the whole length of the secant lines, but the part of the secant line inside the circle. $[b(a + b) = d(c + d)]$

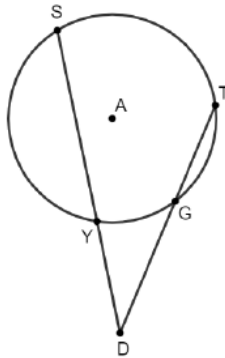
12.2.3 Guided Practice: Solving Problems Involving Intersecting Secant Lengths

Component Overview: Students are guided through using the formula generated in 12.2.2 Exploration to solve for variables and missing lengths. Students are expected to apply this knowledge to real-world situations.



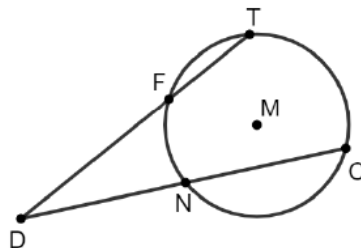
12.2.3 Guided Practice: Solving Problems Involving Intersecting Secant Lengths

- Circle A with secants $\overline{SD}$ and $\overline{TD}$ is shown. $\overline{SD}$ and $\overline{TD}$ intersect at point D outside the circle so that $SY = 5$, $DY = 3$, $TG = x$, and $DG = 4$. Solve for x .



$$3(3 + 5) = 4(4 + x) \quad x = 2$$

- Circle M is shown, where points T , F , N , and C are on circle M . Secants $\overline{TD}$ and $\overline{CD}$ intersect at point D outside the circle such that $TF = 7$, $FD = 9$, $CN = 5x$, and $ND = 4x$. Determine the value of x .



$$4x(9x) = 9(16) \quad x = 2$$



Are you given the lengths to the formula as written? $\overline{DY}(\overline{DY} + \overline{SY}) = \overline{DG}(\overline{DG} + \overline{TG})$. If not, how can you manipulate the lengths you are given to work in this formula?



Some students may need assistance when solving for the value of x . Consider modeling an example for students by sharing your thought process out loud as you solve questions 1 and 2.



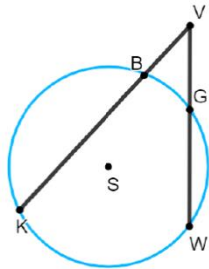
Consider showing students how to color code the information given on the picture using colored pencils or highlighters. This will help students to identify the segment lengths given.

12.2.3 Guided Practice: Solving Problems Involving Intersecting Secant Lengths (continued)

3. Leon Sinks Geologic Area is located in Apalachicola National Forest. The area has 18 sinkholes, many of which create a cave system. A park ranger mapped a circular sinkhole. His diagram



is shown, where circle S has secants $\overline{VK}$ and $\overline{VW}$ such that points B and G are on circle S and point V is outside the circle. $KB = (4x + 4.25)$ meters (m), $VB = 12.75$ m, $GW = 3x + 5.5$ m, and $GV = 14$ m. Find the distance, in meters, from point K to point B .



$$12.75(4x + 4.25 + 12.75) = 14(3x + 5.5 + 14) \quad x = 6.25$$

$$KB = 4(6.25) + 4.25 = 29.25m$$



Allow students to engage in meaningful dialogue related to this real-life situation and challenge students to question the accuracy of this model.

12.2.4 Your Turn!

Component Overview: Students are given the opportunity to take what was practiced in 12.2.3 Guided Practice and apply independently. Probe students as they work using assessing and critical thinking questions to check for understanding, to challenge students, and to help students who are struggling. Consider a peer review process for students to check each other's work as an added layer of support.

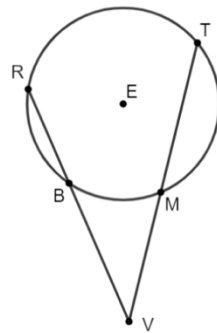


12.2.4 Your Turn!

- Circle E has points R , F , G , and M on the circle. Secants $\overline{RZ}$ and $\overline{MZ}$ intersect at point Z outside the circle. Point G is on $\overline{MZ}$, and point F is on $\overline{RZ}$. $MG = 10$, $GZ = 8$, $RF = (2x - 17)$, and $FZ = 9$. Solve for x .
 $9(2x - 17 + 9) = 8(18) \quad x = 12$
- Tammie was calculating the distance between her favorite places to visit in Epcot. Circle E represents the World Showcase where R , T , M , and B are on the circle and point V is outside the circle at Spaceship Earth. Tammie knows that $RB = 975$ feet (ft), $BV = 1425$ ft, and $VM = 1250$ ft. What is the length of $\overline{MT}$, in feet?

$$1425(1425 + 975) = 1250(1250 + MT)$$

$$MT = 1486 \text{ ft}$$



Encourage students to use a diagram or modeling for making sense of question 1. Provide chart paper for a small group who may be able to engage with question 1 as a group, constructing an Anchor Chart while working.

12.2.5 Wrap-Up: Error Analysis

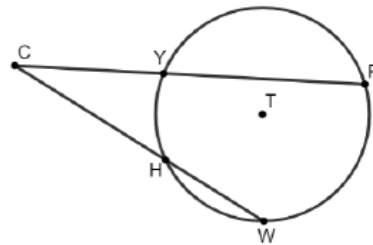
Component Overview: Students complete an Error Analysis by finding the mistake made and writing a note to the students explaining the mistake made when solving for x .



12.2.5 Wrap-Up: Error Analysis

1. Donald was finding the value of x in circle T , where Y , P , W , and H are points on the circle. He knows that $\overline{CP}$ and $\overline{CW}$ are secants of the circle, where $CY = 4$, $YP = x - 3$, $CH = 5$, and $HW = x - 6$. His work is shown.

Donald's Work
$4(x - 3) = 5(x - 6)$
$4x - 12 = 5x - 30$
$x = 18$



Write a note to Donald explaining the mistake he made when solving for x .

Answers will vary.

Donald,

You need to multiply CY by the entire segment length, CP , and multiply CH by the entire segment length, CW , as shown below:

$$4(x - 3 + 4) = 5(x - 6 + 5)$$

$$x = 9$$



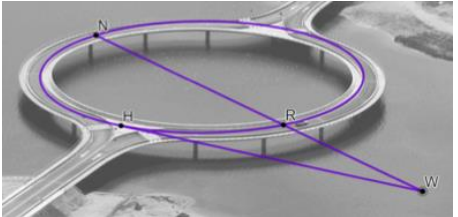
Clarify, Critique, Correct

The intent is to prompt student reflection with an incorrect, incomplete, or ambiguous written argument or explanation, and for students to improve upon the written work by correcting errors and clarifying meaning. This routine fortifies output and engages students in meta-awareness. Consider demonstrating with a think-aloud and press for details when necessary.

12.3 Solving Problems with Secant and Tangent Segments

Lesson Introduction

 Benchmark	MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.		 Learning Target	I can solve mathematical and real-world problems that involve the lengths formed by the intersection of a secant and a tangent segment.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is the relationship of the lengths formed by the intersection of secant and tangent segments? - How can you use the intersection of secant and tangent segments to find the length of missing line segments?			
 Lesson Narrative	This lesson will introduce the students to secant and tangent segments. They will explore the relationship of intersecting secant and tangent lengths, and will work with a partner to write a formula that can be used to find the lengths of missing line segments. Students will be guided through a series of questions, including a question involving a drone flying over Laguna Garzon Bridge in Uruguay. They will conclude by trying questions on their own, including identifying and correcting an error in work.			
 Component Summary	12.3.1 Warm-Up (~5 min.)	Notice and Wonder with patchwork pattern (SE p. 242)	12.3.3 Guided Practice (10-15 min.)	Solving problems involving intersecting secant and tangent lengths (SE p. 244)
	12.3.2 Exploration (10-15 min.)	Discovering intersecting secant and tangent lengths (SE p. 243)	12.3.4 Your Turn! (10-15 min.)	Independent problem solving involving intersecting secant and tangent lengths (SE p. 245)
	12.3.5 Wrap-Up (~1 min.)	Students summarize their learning (SE p. 246)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> - Secant - Tangent		(Optional) Colored Pencils or Markers (Optional) Chart Paper	N/A

 Teacher Tips	Common Misconceptions		Things to Consider
	- Students may be confused between the tangent ratio from trigonometry and the tangent of circles. - Students may confuse multiplying, squaring, and adding in the formula. - Students may confuse a secant line and a tangent line.		Emphasize the point of tangency, and discuss the intersection of the radius to the point of tangency as perpendicular. As you illustrate this theorem, be sure to include a triangle whose sides are the radius of the circle, a line of tangency perpendicular to it, and a segment from the center to a point on the tangent line other than the outer endpoint of the radius.
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns In 12.3.2 Exploration students work with a partner. They take turns, whether it be explaining, justifying, agreeing or disagreeing, or asking clarifying questions. If they disagree, they are expected to support their case and listen to their partner's arguments. The first few times, consider demonstrating with a partner how the discussion is expected to go. Once students are familiar with these structures, less set-up will be necessary. While students are working, ask them to restate their responses more clearly or paraphrase what their partners said.		MA.K12.MTR.4.1: Discussion and Reflection 12.3.2 Exploration allows the students to communicate mathematical ideas, vocabulary, and methods effectively. 12.3.2 Exploration will develop students' ability to justify methods and compare their responses to the responses of their peers.
 Differentiate Instruction	Consider illustrating the theorem on a separate piece of chart paper that you will leave hanging in the classroom as a reference for the students. Students can also co-construct Anchor Charts, either during 12.3.2 Exploration question 1 or the real-world question 3 during 12.3.3 Guided Practice.		
Alternatively, have students use an image or object with a similar shape from the school campus or local community during 12.3.3 Guided Practice, or as an extension opportunity.			
Lesson Notes:			

12.3.1 Warm-Up: Notice and Wonder

Component Overview: Students are shown a Seminole Native American tribal blanket and asked to write down things they notice and wonder about the patchwork on the blanket. The purpose is to make a mathematical task accessible for all students with these low-stakes questions.



12.3.1 Warm-Up: Notice and Wonder

1. The Seminole Native American tribe is known for their novel and colorful dress consisting of patchwork. The decorative technique of patchwork was developed by Seminole women before 1920. Patchwork was rapidly adopted to further embellish the already colorful clothing. As time went on, the designs became more intricate as the seamstresses became more skilled.



- a. What do you notice about the patchwork?
Answers will vary. They use a particular pattern for an entire row. Most of the patterns contain a quadrilateral.
- b. What do you wonder?
Answers will vary. I wonder how they come up with the patterns. Do they have special meaning?



Students can engage in several rounds of Notice and Wonder, so use this as an opportunity to reinforce the expectations and routine of a Notice and Wonder.

12.3.2 Exploration: Lengths of Intersecting Secant and Tangent Segments

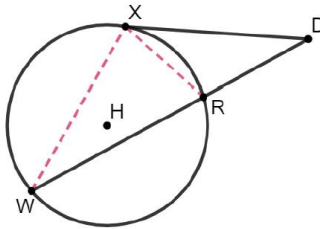
Component Overview: Students are working with intersecting secant and tangent lengths. The outcome of this section is for students to be led into finding the formula for intersecting secant and tangent lines on their own. Students will engage in MA.K12.MTR.4.1 in this collaborative component. Consider giving students space and opportunities to analyze errors and engage in math talk and collaboration.



12.3.2 Exploration: Lengths of Intersecting Secant and Tangent Segments

Work with your partner to complete the following.

- Circle H has points X , R , and W on the circle. Secant $\overline{WD}$ and tangent $\overline{XD}$ intersect at point D outside the circle. It is given that $\triangle DWX \sim \triangle DXR$, $XD = 15$, and $RD = 9$.



- Use the similar triangles to write a proportional relationship that can be used to determine the length of $\overline{RW}$.

$$\frac{XD}{RD} = \frac{WD}{XD}$$

$$\frac{15}{9} = \frac{RW + 9}{15}$$



Notice and Wonder

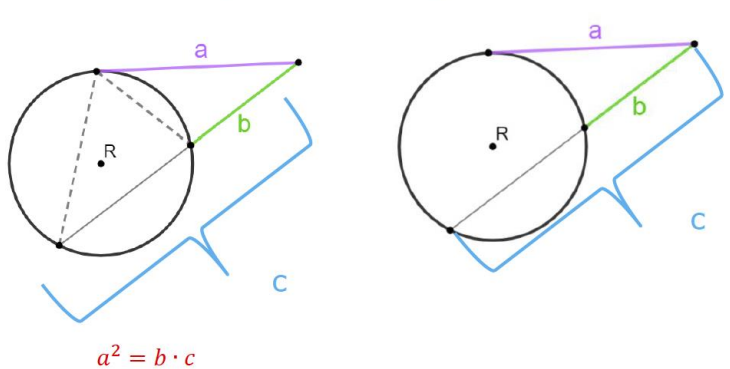
Consider giving students the opportunity to write down things they notice and wonder about the diagrams. This will allow students the chance to make observations before answering the questions.



Students need to pay close attention to the details of the line segment to accurately determine if the line segment is a tangent, secant, or chord before solving questions 1a-b. Encourage precision in responses.

12.3.2 Exploration: Lengths of Intersecting Secant and Tangent Segments (continued)

- b. Use the diagrams of circle R to write a formula that can be used to find the segment lengths of the intersecting secant and tangent lines.



If a secant and tangent segment are drawn to a circle from the same external point, then the square of the measure of the tangent segment is equal to the product of the measures of the secant segment and its external secant segment.

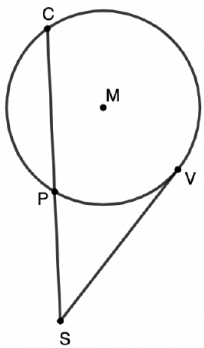
12.3.3 Guided Practice: Solving Problems Involving Intersecting Secant and Tangent Segments

Component Overview: Students are applying their conceptual understanding of intersecting secant and tangent lengths and the formula derived in 12.3.2 Exploration to find the length of line segments.



12.3.3 Guided Practice: Solving Problems Involving Intersecting Secant and Tangent Segments

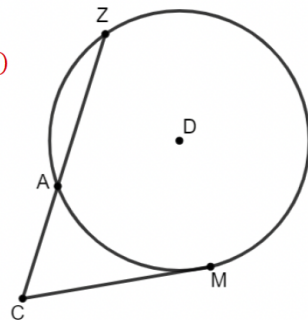
1. Circle M has points C , V , and P on the circle. Secant $\overline{CS}$ and tangent $\overline{VS}$ intersect at point S outside the circle. $CP = 4$ and $SP = 3$. What is the length of $\overline{SV}$? Round to the nearest thousandth.



$$\begin{aligned} SP(SP + CP) &= (SV)^2 \\ 3(7) &= (SV)^2 \\ SV &= 4.583 \end{aligned}$$

2. Circle D is shown with points Z , A , and M on the circle. Secant $\overline{CZ}$ and tangent $\overline{CM}$ intersect at point C outside the circle. $AZ = 16$, $AC = (x - 3)$, and $MC = (x + 3)$. What is the length of $\overline{CM}$?

$$\begin{aligned} (x + 3)^2 &= (x - 3)(x - 3 + 16) \\ x &= 12 \\ CM &= 15 \end{aligned}$$



Use informal observations from 12.3.2 Exploration to inform how to engage students with this activity.

- If student responses sound similar to the blue box in 12.3.2 Exploration, have students lead the class and apply their findings to question 1.
- If students complete 12.3.2 Exploration and struggle making a connection between secants and tangents, use question 1 as a model before students try question 2 in pairs.

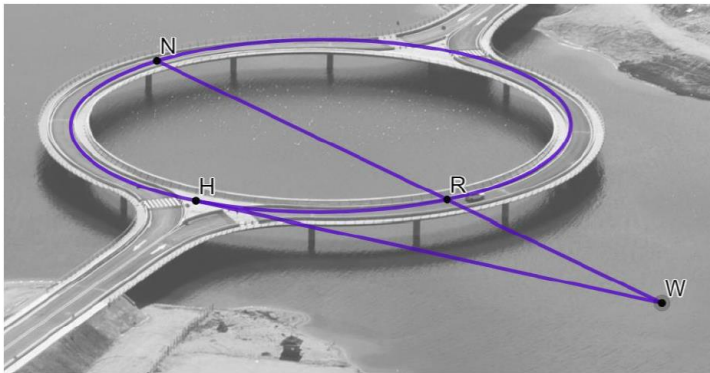


Think-Pair-Share

Allow students time to think about how they would set up the equation. Students can discuss with a partner before sharing with the class. This will ensure students are setting up the equation before solving.

12.3.3 Guided Practice: Solving Problems Involving Intersecting Secant and Tangent Segments (continued)

3. A drone flew over the Laguna Garzon Bridge in Uruguay and took pictures at three points on the bridge: N , R , and H . Tangent $\overline{HW}$ and secant $\overline{NW}$ intersect at point W off the bridge. If $HW = 280$ feet (ft) and $RW = 160$ ft, what is the length of $\overline{NW}$?



$$280^2 = 160(NW), \quad NW = 490 \text{ ft}$$



Allow students to engage in meaningful dialogue related to this real-life situation. Consider using a Notice and Wonder instructional routine. After students have had a chance to think about their responses, ask several students to share things they notice and things they wondered about this question.

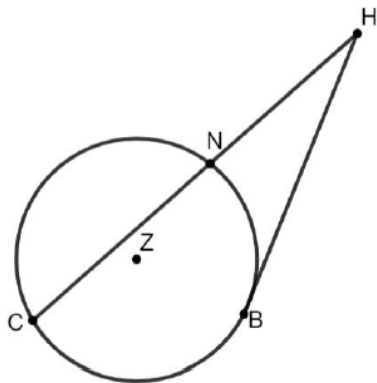
12.3.4 Your Turn!

Component Overview: Students are practicing finding lengths of intersecting secant and tangent and identifying errors made in work.



12.3.4 Your Turn!

1. Circle Z is shown with points C , N , and B on the circle. Secant $\overline{CH}$ and tangent $\overline{BH}$ intersect at point H outside the circle. $CN = (7x - 6)$, $NH = 12$, and $HB = 18$. Find the value of x .



$$18^2 = 12(7x - 6 + 12), \quad x = 3$$



Prompt students to label their diagrams. Students who do not label the diagram and try to solve the problem by just reading the instructions are likely to make mistakes.



Consider asking guiding questions to help struggling students:

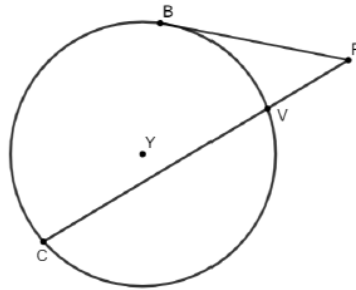
- Which line segment is the tangent?
- What do you know about the length of the tangent?
- Which line segment is the secant?
- What do you know about the length of the secant?



Consider challenging students to engage with question 1 or question 2 without using the image first. This will challenge students to use the information given to identify the error made.

12.3.4 Your Turn! (continued)

2. Isabella and Gregory were working on finding the length of $\overline{BP}$. Circle Y is given, where B , V , and C are points on the circle. Secant $\overline{CP}$ and tangent $\overline{BP}$ intersect at point P outside the circle. $CV = 12$ and $VP = 4$.



MA.K12.MTR.4.1: Engage in discussions that reflect on the mathematical thinking of self and others. Students gain valuable practice for metacognition and analyzing the work of others. Students identify the error in addition to thinking about Gregory's thinking in this work. For students who are unsure how to respond, use prompts like "What do you think Gregory was thinking when he did that? What should he do instead?"

Isabella's and Gregory's work to find the length of $\overline{BP}$ is shown.

Isabella's Work	Gregory's Work
$x^2 = 4(4 + 12)$	$x^2 = 12(4 + 12)$
$x^2 = 4(16)$	$x^2 = 12(16)$
$x^2 = 64$	$x^2 = 192$
$x = 8$	$x = 13.9$

- Whose work do you agree with?
Isabella
- Write a note to the person whose work is incorrect explaining their mistake.
Gregory, you need to find the product of the lengths of the secant segment and its external segment. You found the product of the secant segment and the internal segment.

12.3.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



12.3.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.

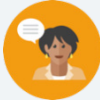


What would reflection look like for you as a teacher? Consider your own scale for strengths and gaps for students through the lesson and how best to fill these gaps. Use this reflection to assess if your students need further clarification of this topic.

12.4 Solving Problems With Lengths of Two Tangents

Lesson Introduction

 Benchmark	MA.912.GR.6.1 Solve mathematical and real-world problems involving length of a secant, tangent, segment, or chord in a given circle.		 Learning Target	I can solve mathematical and real-world problems that involve lengths formed by the intersection of two tangents.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How can you tell the difference between a secant and a tangent line? - What is the relationship between the tangent lines coming from the same point to a circle?			
 Lesson Narrative	Students will explore the relationship between intersecting tangent lines to a circle to discover that two tangent lines that intersect at the same external point are congruent. After discovering the relationship, they will apply that knowledge to solve for missing variables and segment lengths in mathematical and real-world problems.			
 Component Summary	12.4.1 Warm-Up (~5 min.)	Bell Work with secant and tangent lengths (<i>SE p. 247</i>)	12.4.3 Guided Practice (10-15 min.)	Solving problems involving intersecting tangent lines (<i>SE p. 248</i>)
	12.4.2 Exploration (10-15 min.)	Discovery of the tangent segment theorem (<i>SE p. 247</i>)	12.4.4 Your Turn! (10-15 min.)	Applying the tangent segment theorem (<i>SE p. 250</i>)
	12.4.5 Wrap Up (~5 min.)	Summarizing today's lesson in note to absent student (<i>SE p. 251</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> - Secant - Tangent - Congruent		(Optional) Chart Paper (Optional) Markers or Colored Pencils	
			Additional Preparation N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may be confused between the tangent ratio from trigonometry and tangents of circles.	Emphasize the point of tangency and discuss the intersection of the radius to the point of tangency as perpendicular. As you illustrate this theorem, be sure to include a triangle whose sides are the radius of the circle, a line of tangency perpendicular to it, and a segment from the center to a point on the tangent line other than the outer endpoint of the radius.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 12.4.2 Exploration is a collaboration opportunity for students. However, the questions are designed to spark thinking for the individual learner. Allow each student to take time to think about their answers, then share their responses with a partner, and finally with the whole class.	MA.K12.MTR.1.1: Participate in Effortful Learning As students are working, continue to ask them questions that will allow them to analyze the problems in a way that makes sense to them. Encourage students to share thought processes with their partners and ask questions that will help with solving each problem.
 Differentiate Instruction	Consider combining the questions from 12.4.3 Guided Practice and 12.4.4 Your Turn!. Place students in small groups based on their levels of understanding. Each group will work through the questions using the instructional routine Take Turns. Each student will take a turn in completing a step of the question until the answer is found. This will encourage each student to participate in the components and provide students the opportunity to review work and ask clarifying questions, if needed.	
Lesson Notes:		

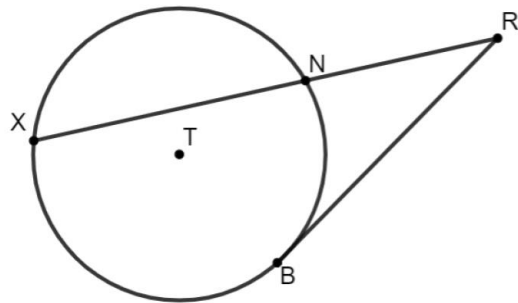
12.4.1 Warm-Up: Bell Work

Component Overview: Students use prior knowledge of the intersection of secant and tangent segments to solve for a missing length.



12.4.1 Warm-Up: Bell Work

- Circle T has points X , N , and B on the circle. Secant $\overline{XR}$ and tangent $\overline{BR}$ intersect at point R outside the circle. Given that $XN = 16$ and $NR = 9$, find the length of $\overline{BR}$.



$$(BR)^2 = 9(25), \quad BR = 15$$



Consider using a Turn and Talk routine here to allow students to reflect and share their thoughts before completing this problem and then discussing whole-group. This allows for students who may struggle to articulate time to practice and verbalize their responses, increasing participation and confidence.

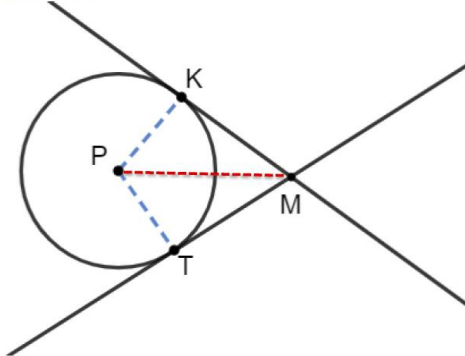
12.4.2 Exploration: Intersecting Tangent Lines

Component Overview: Students explore the relationships between intersecting tangents lines by constructing a triangle to make a connection using prior knowledge of similar triangles. Students work with a partner to reveal that tangent lines to a circle from the same point are congruent.



12.4.2 Exploration: Intersecting Tangent Lines

1. Circle P is shown with tangents $\overline{MK}$ and $\overline{MT}$ and radii $\overline{PK}$ and $\overline{PT}$.



- Draw a dotted line from point P to point M .
See above.
- Discuss with your partner the relationships that can be determined given that $\overline{MK}$ and $\overline{MT}$ are tangents, and $\overline{PK}$ and $\overline{PT}$ are radii.
 $\angle MTP$ and $\angle MKP$ are right angles. Since $PK = PT$ and $PM = PM$, the triangles are congruent by the HL theorem.
- Discuss with your partner why HL congruence can be used to show that $\triangle MKP$ and $\triangle MTP$ are congruent.
- Since $\triangle MKP \cong \triangle MTP$, what is true about $\overline{MK}$ and $\overline{MT}$?
They are congruent.



Tangent segments to a circle from the same point are congruent.



For more precise mathematical responses, consider a Notice and Wonder or Think-Pair-Share for questions 1b and 1c. Alternatively, post prompting questions to guide students to effective exploration:

- What is formed by the line you drew in question 1a?
- What do you know about the relationship between a tangent line and the radius?
- What information do you know about the two triangles?



Listen for phrases like “point of tangency” or similar, and elicit any exemplar resources whole-group to synthesize their findings from 12.4.2 Exploration to apply in 12.4.3 Guided Practice.

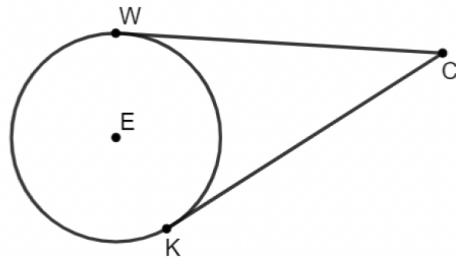
12.4.3 Guided Practice: Solving Problems Involving Intersecting Tangent Lines

Component Overview: Students begin to apply the intersecting tangent theorem in order to find the length of line segments and the value of variables.



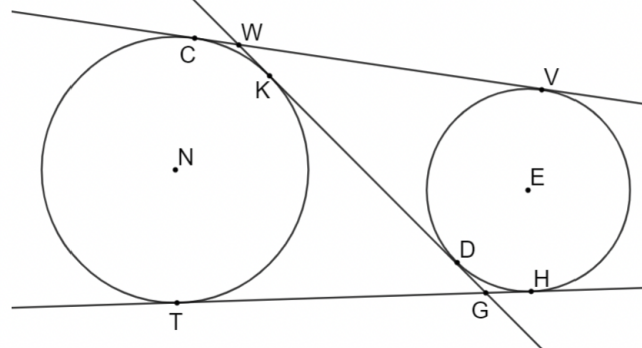
12.4.3 Guided Practice: Solving Problems Involving Intersecting Tangent Lines

- Circle E has points W and K on the circle. Tangents $\overline{WC}$ and $\overline{KC}$ intersect at point C outside the circle. If $WC = x + 15$ and $KC = 2x - 5$, find the length of $\overline{KC}$.



$$x + 15 = 2x - 5, \quad x = 20, \quad KC = 35$$

- Circle N has points C , K , and T on the circle. Circle E has points V , H , and D on the circle. Tangents $\overline{TG}$ and $\overline{KG}$ intersect outside the circle at point G . Tangents $\overline{VG}$ and $\overline{WD}$ intersect outside the circle at point W . If $TG = (7x - 36)$, $KG = (5x - 14)$, $VW = (8y + 13)$, and $WD = KG$, what is the value of y ?



$$5x - 14 = 7x - 36, \quad x = 11, \quad KG = 41$$

$$41 = 8y + 13, \quad y = 3.5$$



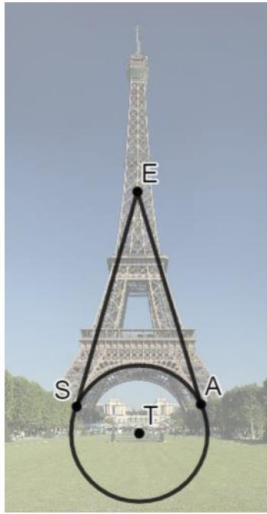
For students who may struggle, consider providing monomial expressions first before students solve a multi-step equation. Also, consider color coding, with colored markers or highlighters, each expression and tangent line for students to see.



What is the relationship between tangent line segments $\overline{TG}$ and $\overline{KG}$? What is the relationship between tangent line segments $\overline{KG}$ and $\overline{VW}$? How can you use these relationships to find the value of y ?

12.4.3 Guided Practice: Solving Problems Involving Intersecting Tangent Lines (continued)

3. The Eiffel Tower in Paris, France stands at 324 meters tall. The tower has four semicircular arches at its base. Circle T represents one of the four arches at the base of the tower, where points S and A lie on the circle. If $SE = (17x + 27)$ meters (m) and $EA = (20x - 15)$ m, what is the length of $\overline{EA}$ in meters?



$$17x + 27 = 20x - 15, \quad x = 14, \\ EA = 265 \text{ m}$$



MA.K12.MTR.6.1: Assess the reasonableness of solutions and MA.K12.MTR.7.1: Apply mathematics to real-world contexts.

Provide opportunity for meaningful reflection with question 3. Applying mathematics to real-work contexts will help connect mathematical concepts to everyday experiences. This question will also challenge students to question the accuracy of the model and the method they used.

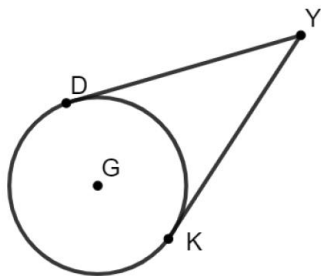
12.4.4 Your Turn!

Component Overview: Students are finding the value of variables by solving questions involving intersecting tangent lines. Students may engage with this component independently or peer review their partner's responses for additional error analysis practice.



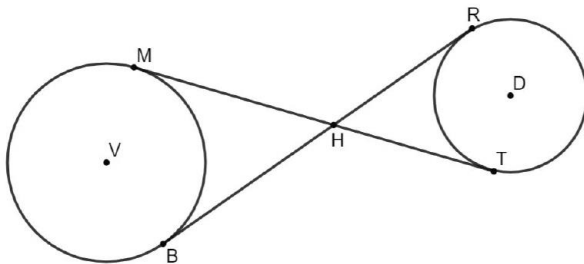
12.4.4 Your Turn!

- Circle G has points D and K on the circle. Tangents $\overline{DY}$ and $\overline{KY}$ intersect at point Y outside the circle. If $DY = 4x - 19$ and $KY = 15$, find the value of x .



$$4x - 19 = 15, \quad x = 8.5$$

- Points M and B are on the circle V and points R and T are on the circle D . $\overline{MT}$ and $\overline{BR}$ intersect at point H . $\overline{MT}$ is tangent to circle V at point M and tangent to circle D at point T . $\overline{BR}$ is tangent to circle V at point B and tangent to circle D at point R . If $MH = (4x - 8)$, $BH = (2x + 1)$, $TH = (2x - 1)$, and $RH = (5y - 7)$, what is the value of y ?



$$2x + 1 = 4x - 8, \quad x = 4.5$$

$$2(4.5) - 1 = 5y - 7, \quad y = 3$$



Clarify, Critique, Correct

Have students exchange their work to analyze the work of their partner. This routine fortifies output and engages students in meta-awareness.

12.4.5 Wrap-Up: Cool Down

Component Overview: Students take time to reflect by writing a short note summarizing the lesson.



12.4.5 Wrap-Up: Cool Down

1. Write a short note summarizing today's lesson to a friend who missed class.

Answers will vary.

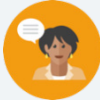


Discuss any questions the students still have regarding the content. Consider a Parking Lot approach with a list of the questions on the board or poster in which students can either answer each other's questions or you can revisit as a class later.

12.5 Pythagorean Theorem and Tangent Lines

Lesson Introduction

 Benchmark	MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.		 Learning Target	I can use the Pythagorean Theorem to solve mathematical and real-world problems involving the length of a tangent segment to a circle.
 Benchmark Coherence	Building On: MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward: N/A	
 Essential Questions	- How can you determine if a line is tangent to a circle? - How can you use the Pythagorean Theorem and tangent lines to calculate line segments?			
 Lesson Narrative	This lesson activates prior knowledge regarding tangent lines to a circle and the Pythagorean Theorem. Students will discover how to utilize the Pythagorean Theorem to determine if a line is tangent to the circle. The primary learning outcome in this lesson focuses on identifying the properties of lines tangent to a circle and using the Pythagorean Theorem and its converse.			
 Component Summary	12.5.1 Warm-Up (~5 min.)	Compose questions about an igloo shelter (<i>SE p. 252</i>)	12.5.3 Guided Practice (10-15 min.)	Using the Pythagorean Theorem and tangent lines (<i>SE p. 254</i>)
	12.5.2 Exploration (15 min.)	Explore Pythagorean Theorem and tangent lines (<i>SE p. 252</i>)	12.5.4 Your Turn! (10 min.)	Individual practice on using the Pythagorean Theorem and tangent lines (<i>SE p. 255</i>)
	12.5.5 Wrap-Up (~5 min.)	Ticket Out the Door demonstrating mastery of learning target (<i>SE p. 256</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Pythagorean Theorem - Tangent - Area - Perimeter		(Optional) Colored Markers or Highlighters	
			Additional Preparation N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may assume all lines tangent to a circle form right angles. - Students may confuse the perimeter and area formulas.	- Students who struggle with the Pythagorean Theorem might need additional support. Consider directing students to a previous lesson, or have a solved Pythagorean Theorem problem available to give them to reference. - Remind students that when finding perimeter and area, they need to include the units with their answers. If the units are not given, instruct them to include the word “units” with their answer.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Clarify, Critique, Connect Consider implementing this routine by allowing students to trade their response in 12.5.3 Guided Practice and 12.5.4 Your Turn! with another student. Students will then ask their partner clarifying questions, as needed. This process will allow students to think more analytically and provide a more detailed mathematical explanation.	MA.K12.MTR.3.1: Mathematical Fluency Mathematical fluency involves being accurate, efficient, and precise while carrying out calculations. By providing students with multiple opportunities to execute procedures and perform calculations, this lesson allows students to accurately determine the length of line segments and determine if a line is tangent to a circle.
 Differentiate Instruction	Read and discuss questions aloud to assist students who benefit from an auditory entry-point. Encourage students to use the diagrams to help make sense of each question. Consider showing students how to color code each expression and tangent line with colored markers or highlighters. To provide a kinesthetic entry-point, consider placing the questions from 12.5.3 Guided Practice and 12.5.4 Your Turn! around the room to create a Gallery Walk.	
Lesson Notes:		

12.5.1 Warm-Up: Ice Problem

Component Overview: Students are shown a picture of an igloo and are asked to write specific questions related to the image.



12.5.1 Warm-Up: Ice Problem

1. An igloo, also known as a snow house or snow hut, is a type of shelter built out of snow.



- a. What is the first question that comes to your mind?
Answers will vary. It is warm? How do they stay warm? Why is the snow house built like a sphere instead of a block?
- b. Write a question that would require someone to use geometry to solve.
Answers will vary. If the circular part of the igloo has a radius of 5 feet (ft), can you find the approximate area of the igloo given it is not a full circle?



This question provides the opportunity for interdisciplinary studies. Consider challenging students by having them incorporate information about igloos to lessons they have studied in their Civics or other Social Studies classes. You could also use the science of how an igloo retains its heat as an extension activity.

12.5.2 Exploration: Pythagorean Theorem and Tangent Lines

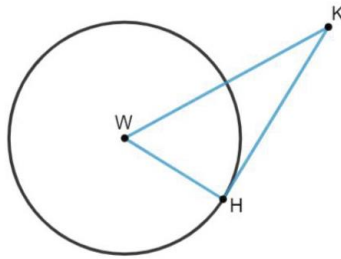
Component Overview: Students explore the relationship between tangent lines of a circle and the Pythagorean Theorem. Students will have the opportunity to collaborate with a partner and other classmates regarding their conclusions.



12.5.2 Exploration: Pythagorean Theorem and Tangent Lines

Work with your partner to complete the following.

- Circle W has point H on the circle. $\overline{WK}$ and tangent $\overline{KH}$ intersect at point K outside the circle.



- Discuss with your partner what can be concluded from the given information. Justify your conclusions. Give at least two conclusions.
Answers will vary. $\overline{WH}$ is a radius because point W is the center of the circle and point H lies on the circle. $\angle KHW$ is a right angle because a tangent is perpendicular to a radius at the point of tangency.



Think-Pair-Share

Consider providing students with the opportunity to use this instructional routine prior to answering the questions. This will allow students to make observations and conclusions in a low-stakes way before sharing with a partner and/or class.

12.5.2 Exploration: Pythagorean Theorem and Tangent Lines (continued)

- b. Ask a classmate for two conclusions they drew. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	My Summary of Their Reason	Initials
<i>Answers will vary.</i>		

- c. Discuss with your partner what information is needed to determine the length of $\overline{WK}$.
The length of $\overline{HW}$ and $\overline{KH}$.
- d. Explain why any triangle drawn, where the radius and a tangent are the legs, will result in a right triangle.
If a line is tangent to a circle and perpendicular to the radius at the point of tangency, it will form a right angle.

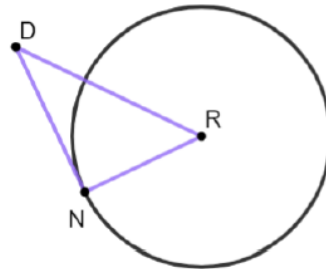


Consider prompting students with questions before answering question 1d:

- What is the relationship between a tangent line and the radius of a circle?
- Will this always be true? Why or why not?
- Can you create an example where the radius and a tangent line will not result in a right triangle?

12.5.2 Exploration: Pythagorean Theorem and Tangent Lines (continued)

2. Circle R is shown with point D outside the circle, where point N is on the circle. $DN = 10$, $NR = 8$, and $DR = 4\sqrt{10}$.



Students may believe that the given line segment is automatically tangent, therefore creating a right angle and right triangle. During question 2, remind students that in Geometry, not all diagrams are drawn to scale and that we cannot assume something is true without completing the math to prove it is true.

- a. Use the Pythagorean Theorem to determine if $\overline{DN}$ is a tangent to circle R .

$$10^2 + 8^2 = 2\sqrt{41}, \quad 2\sqrt{41} \neq 4\sqrt{10}$$

- b. Complete the statements.

$\triangle DRN$ ☐ is
☒ is not a right triangle. Therefore, $\overline{DN}$

☐ is
☒ is not a tangent line to circle R .

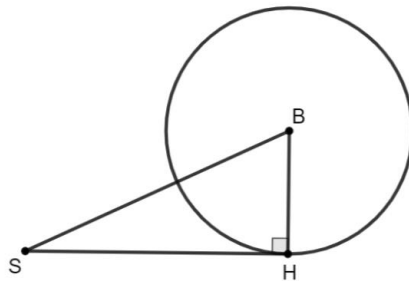
12.5.3 Guided Practice: Pythagorean Theorem and Tangent Lines

Component Overview: Students will use their knowledge of the Pythagorean Theorem and tangent lines to prove a line is tangent to a circle, and to find lengths of line segments. Students will use the lengths of line segments found to calculate the areas and perimeters of triangles.



12.5.3 Guided Practice: Pythagorean Theorem and Tangent Lines

1. Circle B has point H on the circle and point S outside the circle.
 $m\angle SHB = 90^\circ$,
 $SB = 13$, and
 $BH = 5$.



- a. What is the length of $\overline{SH}$?

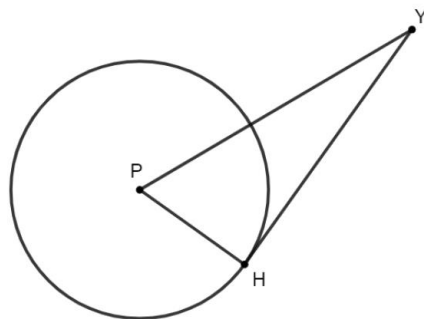
$$SH^2 + 5^2 = 13^2, \quad SH = 12$$

- b. What is the area of $\triangle SBH$?

$$\frac{1}{2}(5)(12) = 30 \text{ units squared}$$

2. Circle P has point H on the circle and point Y outside the circle, where $\overline{PY}$ and tangent $\overline{HY}$ intersect. $PH = 10$, and $HY = 17$.

What is the length of $\overline{PY}$?



$$10^2 + 17^2 = PY^2, \quad PY = \sqrt{389}$$



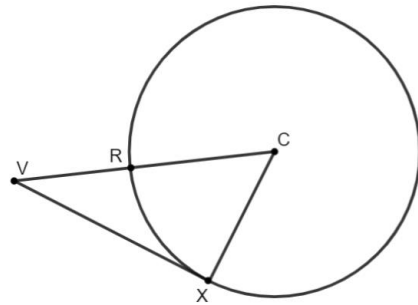
Use inquiry to guide students through question 1 before they try questions 2 and 3 on their own:

- How can you determine that a line is tangent to a circle?
- What specific information do you need for this?
- What is the difference between a tangent line and a secant line?
- How can tangent lines of a circle help you to find the area and perimeter of a triangle?

12.5.3 Guided Practice Pythagorean Theorem and Tangent Lines (continued)

3. Circle C has points R and X on the circle and point V outside the circle, where $\overline{VC}$ and tangent $\overline{VX}$ intersect at point V . $VX = 24$ and $VR = 16$.

- a. What is the length of $\overline{CX}$?



Let x equal the length of the radius, $\overline{CR}$ and $\overline{CX}$.

$$(16 + x)^2 - 24^2 = x^2, \quad x = 10$$

$$CX = 10$$

- b. What is the length of $\overline{VC}$?

$$VC = 26$$

- c. What is the perimeter of $\triangle VCX$?

$$\text{Perimeter} = 60 \text{ units}$$



Provide the question stems that were used in question 1 for students to use on their own as additional scaffolding for question 3. Alternatively, prompt students to use the Study Expert video for hints while working on question 3.

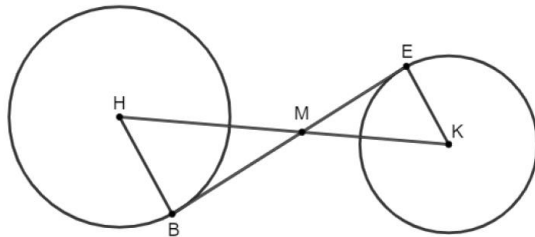
12.5.4 Your Turn!

Component Overview: Students work independently to use tangent lines to find the perimeter and area of triangles, and to identify an error in work.



12.5.4 Your Turn!

1. Circle H has point B on the circle and circle K has point E on the circle. $\overline{HK}$ and tangent $\overline{EB}$ intersect outside the circles at point M . $m\angle HBM = m\angle MEK = 90^\circ$, $HB = 6$, $BM = 8$, and $MK = 8$.



- a. Explain why $\triangle HMB \sim \triangle KME$.

Since $\angle HBM = \angle MEK$, they are congruent to each other. $\angle HMB \cong \angle KME$ by the vertical angles theorem. Therefore, the two triangles are similar by AA similarity.

- b. What is the length of $\overline{HK}$?

$$6^2 + 8^2 = HM^2, \quad HM = 10$$

$$HM + MK = HK, \quad HK = 18$$

- c. What is the perimeter of $\triangle KME$?

$$\frac{HM}{MK} = \frac{HB}{EK}, \quad EK = 4.8$$

$$\frac{HM}{MK} = \frac{BM}{EM}, \quad ME = 6.4$$

$$\text{Perimeter of } \triangle KME = 19.2 \text{ units}$$

- d. What is the area of $\triangle KME$?

$$\frac{1}{2}(EK)(ME) = 15.36 \text{ square units}$$



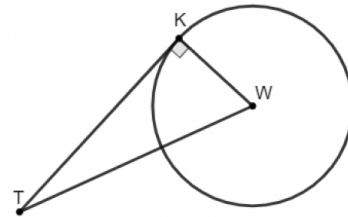
Students may struggle with how to find the perimeter in question 1. Consider reminding students how they can use proportions to determine if triangles are similar, and then how to apply this information to solve question 1c.



Students who see this image as a bicycle may benefit from visualizing the dimensions as a bicycle to find the perimeter.

12.5.4 Your Turn! (continued)

2. Grace and Zackary are finding the length of tangent $\overline{TK}$ of circle W , where K is a point on the circle. $\overline{TW}$ and tangent $\overline{TK}$ intersect at point T outside the circle. $TW = 18$, and $KW = 7$.



Grace's and Zackary's work is shown.

Grace's Work	Zackary's Work
$7^2 + x^2 = 18^2$	$7^2 + 18^2 = x^2$
$49 + x^2 = 324$	$49 + 324 = x^2$
$x^2 = 275$	$x^2 = 373$
$x = 5\sqrt{11}$	$x = \sqrt{373}$

Write a note to the person whose work is incorrect explaining their mistake.

**Means
I
Start
To
Acquire
Knowledge
Experience
Skills**

Answers will vary. Zackary, you were on the right track to recognize that you can use the Pythagorean Theorem to find the length of tangent $\overline{TK}$. However, $\overline{TW}$ is the hypotenuse of the right triangle, not a leg. Next time, it may help to label the triangle first to ensure you substitute the values in for the correct variables.



MA.K12.MTR.4.1: Engage in discussions that reflect on the mathematical thinking of self and others. Provide opportunity for students to get creative with their response. Some students might prefer to write a note, while other students might better engage or activate agency by making a video using technology, or social media.

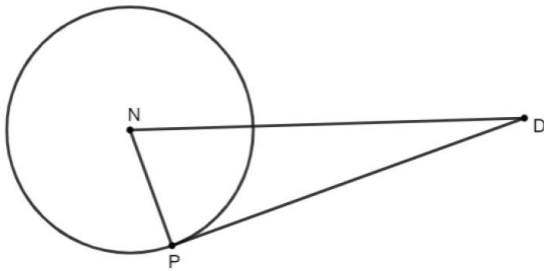
12.5.5 Wrap-Up: Ticket Out the Door

Component Overview: Students find the length of a line segment to reflect their level of understanding of the content of this lesson.



12.5.5 Wrap-Up: Ticket Out the Door

- Circle N has point P on the circle and point D outside the circle at the intersection of $\overline{DN}$ and tangent $\overline{DP}$.
 $NP = 2$ and $DP = 6$.



What is the length of $\overline{ND}$?

$$2^2 + 6^2 = ND, \quad ND = 2\sqrt{10}$$



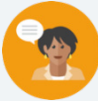
Consider identifying specific strands from 12.5.5 Wrap-Up to inform small groups in Lesson 6:

- Identifying key measurements
- Labeling the diagram
- Explanation of how you found $\overline{ND}$

12.6 Lengths of Secants, Tangents, Segments, and Chords in Circles

Lesson Introduction

 Benchmark	MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.		 Learning Target	I can solve mathematical and real-world problems involving the length of segments of a circle formed by intersections.
 Benchmark Coherence	Building On: N/A		Working Toward: N/A	
 Essential Questions	What is the difference between a chord, secant, and tangent line to a circle?			
 Lesson Narrative	This lesson will bring together all the properties of chord, secant, and tangent lines in a circle. During this lesson, students will be matching each description with the correct image, type of line segment, and answer. Students will need to be able to differentiate between the types of line segments of circles in order to use the correct method of solving.			
 Component Summary	12.6.1 Warm-Up (~5 min.)	Consider how math is a mode of communication (SE p. 257)	12.6.3 Try It (10-15 min.)	Real-world problems solving for length (SE p. 259)
	12.6.2 Activity (30-35 min.)	Matching activity with line segments of circles (SE p. 258)	12.6.4 Wrap-Up (~5 min.)	Students summarize their learning (SE p. 260)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Chord - Secant - Tangent		- Match Sheet Activity Pages (Optional) Poster or Chart Paper	If you want to pre-cut the math sheet activity pages, students could group them together on their desk for each question.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse the difference between a chord, secant, and tangent line in a circle. - Students may forget the different formulas to solve to find a chord, secant, or tangent line in a circle. - Students may miscalculate when using the formulas, leading them to an incorrect answer.	Consider having all the different line segments of a circle with each corresponding formula and method of solving posted throughout your room for students to use as a reference.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 12.6.2 Activity is a card matching activity. Students have quiet time to think about the activity and work on it individually. Then, they take time to share their responses or their progress with a partner. If time allows, once these partner conversations have taken place, some partnerships can be selected to share their thoughts with the class.	MA.K12.MTR.3.1: Mathematical Fluency This lesson is an opportunity for students to complete several tasks with mathematical fluency. They will need to select efficient and appropriate methods for solving these questions, given the different lines of the circle. The goal of the lesson is for them to be able to complete the different question types with accuracy and confidence.
 Differentiate Instruction	12.6.2 Activity is meant to be completed individually. If you anticipate that students might struggle during this activity, consider having the different types of line segments of circles posted in your room on posters and have the students do a Gallery Walk before starting this lesson. At each type of line segment of circles, have students write down on the poster things they know about the line segments, examples, and/or drawings. Provide a visual entry-point from seeing the connection between the description, type of line segment, and the image. Provide an auditory entry-point from the collaborative conversation at each poster. Provide a kinesthetic entry-point from moving around the room from poster to poster.	
Lesson Notes:		

12.6.1 Warm-Up: Critical Thinking

Component Overview: Students are asked to use critical thinking to answer questions.



12.6.1 Warm-Up: Critical Thinking

1. Explain how math could be considered a language used for communication.

Answers may vary

*Math, like other languages, is a method of communicating.
It consists of structure, symbols, and rules.*

2. Suppose you were taking Geometry in another country. How do you think the problems would differ?

Answers may vary.

The vocabulary would be different.



Do you have students who speak other languages in your classroom? Consider inviting a guest speaker who speaks a foreign language to join class (in person or virtually). Additionally, consider having students return to this lesson later in the year for project-based learning around this topic.

12.6.2 Activity: Line Segments of Circles

Component Overview: Students tie together the different line segments of circles studied in Unit 6, including: chords, secant lines, and tangent lines.



12.6.2 Activity: Line Segments of Circles

Match each image, type of segment, and answer to the description of circles. Record your answers in the table.

Circle	Work	Cards' Letter
Circle K has points G , N , D , and E on the circle, where $GM = (2x + 2)$, $MD = 14$, $EM = 12$, and $MN = (2x + 5)$	$(2x + 2)(14) = (2x + 5)(12)$ $x = 8$	Image: <u>W</u> Segment: <u>E or D</u> Answer: <u>M</u>
Circle H has points C , T , and B on the circle, where $BR = 15$, $RT = 9$, and $CT = x$	$15^2 = 9(x + 9)$ $x = 16$	Image: <u>N</u> Segment: <u>F</u> Answer: <u>K</u>
Circle R has points T , S , V , and W on the circle, where $TS = (x - 8)$, $SP = 4$, $VP = 3$, and $VW = (x - 5)$	$(x - 8 + 4)(4) = (x - 5 + 3)(3)$ $x = 10$	Image: <u>P</u> Segment: <u>B</u> Answer: <u>H</u>
Circle F has points E and B on the circle, where $EY = (7x - 3)$ and $BY = (5x + 1)$	$7x - 3 = 5x + 1$ $x = 2$	Image: <u>R</u> Segment: <u>A</u> Answer: <u>G</u>
Circle V has point B on the circle, where $VB = 4$, $BA = 8$ and $VA = x$	$4^2 + 8^2 = x^2$ $x = 4\sqrt{5}$	Image: <u>S</u> Segment: <u>C</u> Answer: <u>L</u>
Circle A has points G , R , H , and S on the circle, where $MA = WA$, $GR = 35$, and $SH = (6x - 7)$	$6x - 7 = 35$ $x = 7$	Image: <u>T</u> Segment: <u>E or D</u> Answer: <u>J</u>



Group Presentation

Consider assigning each group one of the problems in 12.6.2 Activity. Each group will present to the class by explaining which line segments are used, which image matches the circle, and justifying the answer. Students can use this time to ask clarifying questions, and ask members of each group to restate their understanding of the group's findings in their own words.



Consider challenging students by removing the answer cards. Students can verify their answers at the end with a partner.

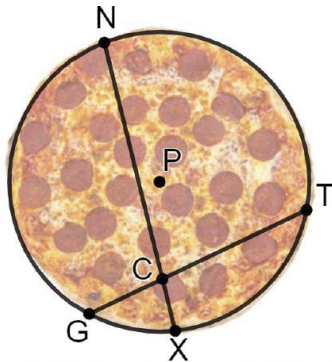
12.6.3 Try It: Solving for Length

Component Overview: This component positions students to make connections between chords, secants, and tangent lines in mathematical and real-world contexts.



12.6.3 Try It: Solving for Length

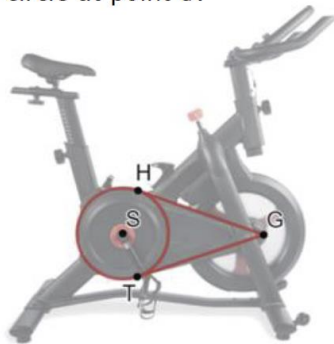
- When Doug's pizza was delivered, he realized the pizza was not cut evenly. The pizza, circle P , is cut by chords NX and GT .



Given that $NC = (5x - 6)$ inches (in), $CX = 2$ in, $GC = 3$ in, and $CT = (2x)$ in, what is the length of NX ?

$NX = 11$ in.

- A belt guard is used on stationary bikes to cover some of the inner mechanics. Circle S is shown, where points H and T are on the circle. Tangents HG and TG intersect outside the circle at point G .



If $HG = 15$ and $TG = 4y - 9$, what is the value of y ?

$y = 6$



Use questioning techniques and methods to create a student-centered learning environment in which students are driving the learning. Consider using sentence stems to encourage conversations:

- "I had to pay attention to how I labeled the diagram because..."
- "I know the line segments in this question are... because..."
- "I needed to make sure I was using the correct formula because..."

12.6.4 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



12.6.4 Wrap-Up: Cool Down

1. Write **one question** you still have about solving mathematical and real-world problems involving the lengths of secants, tangents, segments, or chords of a circle.

Answers may vary.



Lessons 7-9 explore the equation of a circle. Consider utilizing EdgeXL for a summative assessment of Lessons 1-6 to determine student progress regarding secants, tangents, and problem-solving with circles.

12.7 Deriving the Equation of a Circle

Lesson Introduction

 Benchmark	MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.		 Learning Targets	- I can use the Pythagorean Theorem to derive the equation of a circle. - I can determine key features of a circle given the graph or equation. - I can write the equation of a circle using key features.	
 Benchmark Coherence	Building On:			Working Toward:	
	MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.			N/A	
 Essential Questions	- How can you use the Pythagorean Theorem to derive the equation of a circle? - What are the two key features of a circle? - How can you write the equation of a circle when given the center and radius?				
 Lesson Narrative	This lesson activates prior knowledge regarding the Pythagorean Theorem. While using the Pythagorean Theorem, students will use the key features of a circle to derive the equation of a circle in the center-radius form. This will be done in the coordinate plane where students are given coordinates to find distance. Students will have the opportunity to use the formula derived to write the equation of a given circle on a coordinate plane.				
 Component Summary	12.7.1 Warm-Up (~5 min.)	Notice and wonder with Roundabouts (<i>SE p. 261</i>)	12.7.3 Guided Instruction (10-15 min.)	Using key features to write the equation of a circle (<i>SE p. 265</i>)	
	12.7.2 Exploration (25-30 min.)	Deriving the equation of a circle (<i>SE p. 261</i>)	12.7.4 Wrap-Up (~5 min.)	Ticket Out the Door determining key features and writing equation of circle (<i>SE p. 266</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Center-Radius Form <u>Additional Terms:</u> - Pythagorean Theorem		N/A		The Pythagorean Theorem is used extensively in this lesson. Consider having reference cards available for struggling students who are having trouble with remembering the theorem.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle finding the length of the segment when given variables instead of numbers. - Students may have forgotten which part of the triangle are the legs and which part is the hypotenuse. - Students may believe that distance can be negative.	12.7.2 Exploration is meant to derive the equation of the circle, not necessarily find the actual length of a segment. - Pay close attention to the circles whose center coordinates are a combination of both positive and negative values and guide students through the process of connecting the equation of the circle to the sign of the numbers.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time 12.7.2 Exploration is structured as an interactive opportunity for students to revise and refine both their ideas and their verbal and written output. This routine allows students to ask clarifying questions, if needed, and continue on to the next question. Consider having students do this throughout the component, so students can get stronger with writing equations of a circle in general form.	MA.K12.MTR.1.1: Participate in Effortful Learning 12.7.2 Exploration actively allows students to participate in effortful learning, both individually and collectively. Students will analyze the problems, ask questions that will help them find the equation of the circle, and help and support each other while attempting the next problems in the set. 12.7.2 Exploration will foster perseverance in students as they solve progressively more difficult problems.
 Differentiate Instruction	12.7.2 Exploration is set up to allow students to work through a series of questions that become progressively more complex. The diagrams of the circles referenced throughout 12.7.2 Exploration can assist student who need a visual of how the Pythagorean Theorem and writing the equation of a circle are related. There is also a lot of reading in this section. Provide an auditory entry-point from hearing the verbal description of the written representations.	
Lesson Notes:		

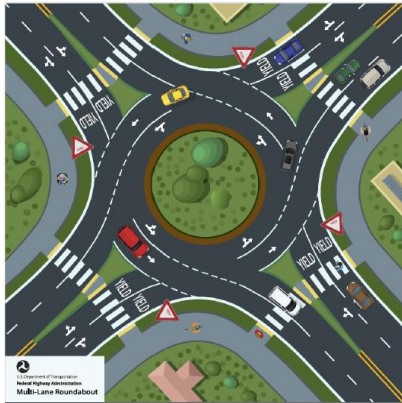
12.7.1 Warm-Up: Roundabouts

Component Overview: Students examine the concept of circles using a real-life example of roundabouts.



12.7.1 Warm-Up: Roundabouts

Roundabouts promote a continuous flow of traffic because drivers do not have to wait for a green light at a roundabout to get through the intersection. A diagram of a roundabout is shown.



of



How are tangent lines similar to the far-right lanes entering the roundabout?

How could circumference be used when finding the distance a car traveled through a roundabout?

1. What do you wonder about the diagram?

Answers may vary.

I wonder how traffic is controlled in a roundabout. I wonder why there are no cars in the inner circle of the roundabout.

2. How many different paths could the red car take?

The red car can only take one path because he is in the outer circle.

3. List at least two things you have studied in Geometry so far that may be helpful to know when constructing a roundabout.

Answers may vary.

What is the distance around the roundabout?

How many feet across is the roundabout?

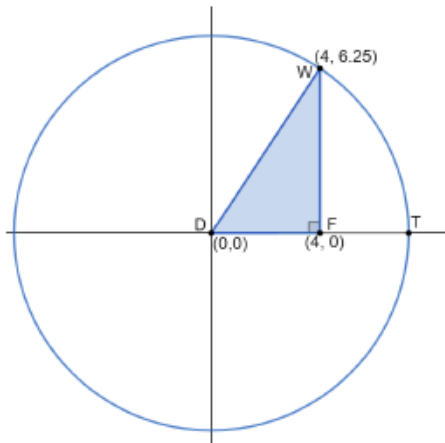
12.7.2 Exploration: Deriving the Equation of a Circle

Component Overview: Students use diagrams to derive the equation of a circle on the coordinate plane. Students use given coordinates and the Pythagorean Theorem to find the side lengths of the triangle created in a circle, which will lead students into writing the equation of a circle.



12.7.2 Exploration: Deriving the Equation of a Circle

1. The center of circle D is located at $(0,0)$. Point W lies on circle D at the point $(4, 6.25)$. Right triangle DFW is drawn inside the circle, such that DW is a circle's radius and point F , $(4,0)$, lies on radius $\overline{DT}$.



- a. Find the length of each side.

Line Segment	Distance Between Endpoints
$\overline{DF}$	4
$\overline{FW}$	6.25

- b. Use the values you found for the lengths of $\overline{DF}$ and $\overline{FW}$ to complete the equation that can be used to determine the length of $\overline{DW}$ by applying the Pythagorean Theorem.

$$(DW)^2 = (4)^2 + (6.25)^2$$



Compare and Connect

Begin and end 12.7.2 Exploration by prompting students to make connections between what they know about circles, and what they know about their equations.

Before the beginning: What do you know about the equation of a circle? Take a guess – what do you think it looks like? Why?

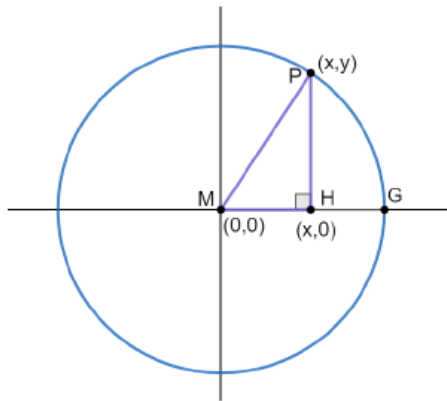
After the activity: What do you notice? What did you discover? How would you describe the relationship between a circle and its equation? How is that represented with its graph?



As 12.7.2 Exploration is meant to derive the equation of the circle, it is not necessary for the students to determine the actual length.

12.7.2 Exploration: Deriving the Equation of a Circle (continued)

2. The center of circle M is located at $(0,0)$. Point P lies on circle M at the point (x,y) . Right triangle MPH is drawn inside the circle, such that $\overline{MP}$ is a circle's radius and point H , $(x,0)$, lies on radius $\overline{MG}$.



How is this circle different from the circle in question 1? How is finding the distance of each line segment similar? How is it different?

- a. Write an expression that represents the length of each side.

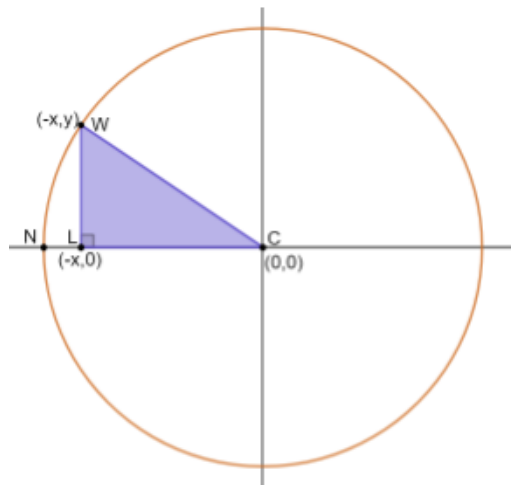
Line Segment	Distance Between Endpoints
$\overline{MH}$	x
$\overline{HP}$	y

- b. Use the expressions you found for the lengths of $\overline{MH}$ and $\overline{HP}$ to complete the equation that can be used to determine the length of $\overline{MP}$ by applying the Pythagorean Theorem.

$$(MP)^2 = (\quad x \quad)^2 + (\quad y \quad)^2$$

12.7.2 Exploration: Deriving the Equation of a Circle (continued)

3. The center of circle C is located at $(0,0)$. Point W lies on circle C at the point $(-x,y)$. Right triangle WLC is drawn inside the circle, such that WC is a circle's radius and point L , $(-x,0)$, lies on radius NC .



Students may need to be reminded that distance is always positive. Consider prompting students to engage the class in a discussion:

Can distance be negative? Why or why not?

- a. Write an expression that represents the length of each side.

Line Segment	Distance Between Endpoints
LC	x
LW	y

- b. Use the expressions you found for the lengths of LC and LW to complete the equation that can be used to determine the length of WC by applying the Pythagorean Theorem.

$$(WC)^2 = (x)^2 + (y)^2$$

12.7.2 Exploration: Deriving the Equation of a Circle (continued)

- c. Complete the table with the equation you found that can be used to determine the length of the radius of each circle.

Line Segment	Distance
$\overline{MP}$ radius of circle M	$(MP)^2 = x^2 + y^2$ or $(r)^2 = x^2 + y^2$
$\overline{WC}$ radius of circle C	$(WC)^2 = x^2 + y^2$ or $(r)^2 = x^2 + y^2$

In circle M and circle C , the location of the point on the circle was in a different location, but the equations that can be used to determine the length of the radius of the circles are the same.

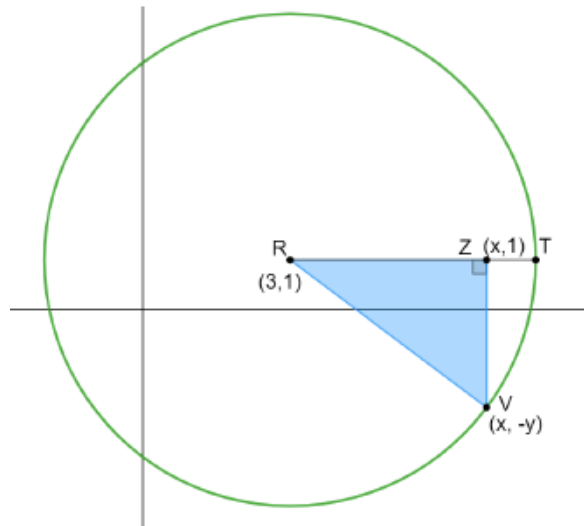


Use inquiry to drive learning. Listen for students who are stuck or students who need support making connections.

- What happens when you square a number?
- What happens when you square a negative number?
- Is there a difference between -5^2 and $(-5)^2$?
- Why is this difference important in deriving the equation of a circle?

12.7.2 Exploration: Deriving the Equation of a Circle (continued)

4. The center of circle R is located at $(3,1)$. Point V lies on circle R at the point $(x, -y)$. Right triangle RZV is drawn inside the circle, such that $\overline{RV}$ is a circle's radius and point $Z, (x, 1)$, lies on radius $\overline{RT}$.



- a. Write an expression that represents the length of each side.

Line Segment	Distance Between Endpoints
$\overline{RZ}$	$x - 3$
$\overline{ZV}$	$1 + (-y)$

- b. Complete the equation that can be used to determine the length of the radius of circle R .
 $(r)^2 = (\quad x - 3 \quad)^2 + (\quad y - 1 \quad)^2$



Question 4 allows students to investigate equivalent forms of the circle. Consider investigating the different ways students can represent the distance and then explain why the value squared would be the same.

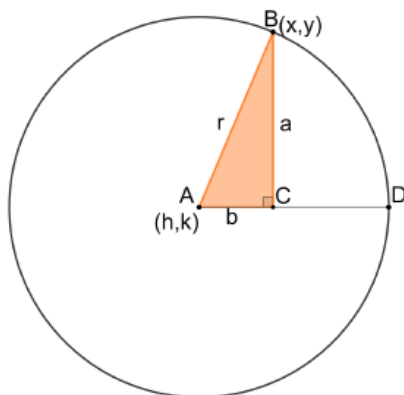


Question 4 provides an opportunity to determine students' understanding of the key elements of a circle and points in the coordinate plane.

Students who struggle with finding the distance in this problem might need you to provide numbers for some of the variables until they can see the relationship.

12.7.2 Exploration: Deriving the Equation of a Circle (continued)

5. The center of circle A is located at (h, k) . Point B lies on circle C at the point (x, y) . Right triangle ABC is drawn inside the circle such that $\overline{AB}$ is a circle's radius and point C , lies on radius $\overline{AD}$. $AC = b$, $BC = a$, and $AB = r$.



- a. Use (x, y) and (h, k) to determine the ordered pair for point C .
 (x, k)
- b. Fill out the table by using the ordered pairs to complete each equation.

Line Segment	Distance Between Endpoints
$\overline{AC}$	$b = x - h$
$\overline{BC}$	$a = y - k$

The Pythagorean Theorem can be used to write the equation, $a^2 + b^2 = r^2$.

- c. Use the equations in the table to substitute for a and b in the equation $a^2 + b^2 = r^2$.
 $(x - h)^2 + (y - k)^2 = r^2$



The **center-radius form** of an equation of a circle with center, (h, k) , and radius, r , is $(x - h)^2 + (y - k)^2 = r^2$.



Students may struggle with question 5. Refer them to the other circles to determine the relationship.

- Why is the x –value of the ordered pair the same as point A ?
- Why is the y –value of the ordered pair the same as point B ?



Consider having students discuss what they have learned in 12.7.2 Exploration with a partner or small group using the Compare and Connect instructional routine. Students are encouraged to share comparisons found, make connections between each equation they wrote throughout this component, and discuss how the equation of a circle comes from it.

12.7.3 Guided Instruction: Key Features of a Circle

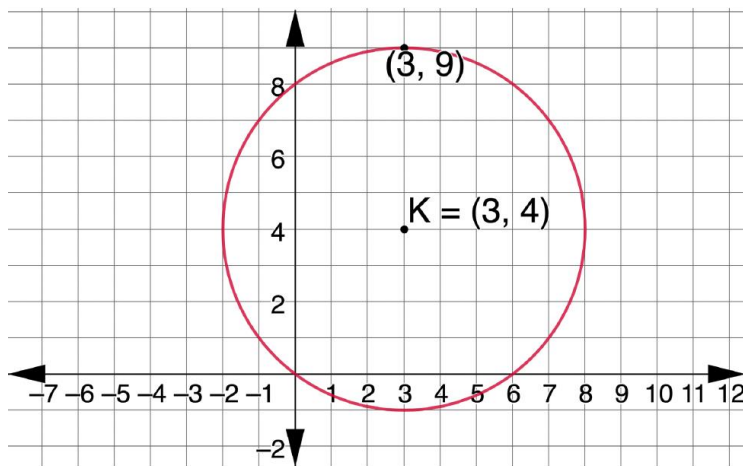
Component Overview: Students are using the equation of a circle derived in 12.7.3 Exploration to write the equation of a circle when given a center and radius.



12.7.3 Guided Instruction: Key Features of a Circle

Circles have two key features: the center and the radius. The key features can be used to write the equation, $(x - h)^2 + (y - k)^2 = r^2$, of the circle.

1. Circle K is shown with center $(3, 4)$. The point $(3, 9)$ is on circle K .



- a. Determine the length of circle K 's radius.
 $r = 5$
- b. Write the equation that represents circle K .
 $(x - 3)^2 + (y - 4)^2 = 25$ or
 $(x - 3)^2 + (y - 4)^2 = 5^2$

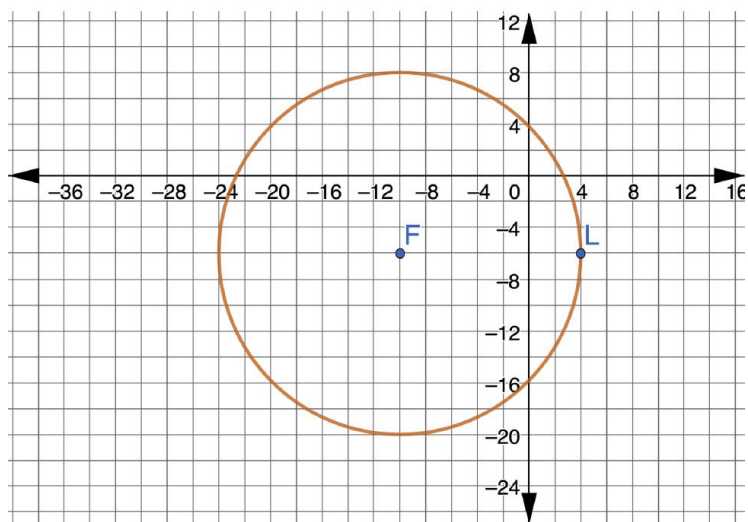


Notice and Wonder

Consider providing students the opportunity to write down things they notice and wonder about the equation of the circle and the circle provided in question 1. This will allow students to make connections between the equation and the information given.

12.7.3 Guided Instruction: Key Features of a Circle (continued)

2. Circle F is shown, where point L is on the circle.



Provide opportunity for synthesis using question 2. Students spent time exploring in the first half of this lesson, and the next activity is a formative Wrap-Up. Utilize question 2 to ask questions that position students as sensemakers to synthesize their learning to apply during 12.7.4 Wrap-Up.

- a. Determine the key features of circle F .

Key Features	
Radius	14
Center	$(-10, -6)$

- b. Write the equation that represents circle F .

$$(x + 10)^2 + (y + 6)^2 = 196 \text{ or}$$

$$(x + 10)^2 + (y + 6)^2 = 14^2$$

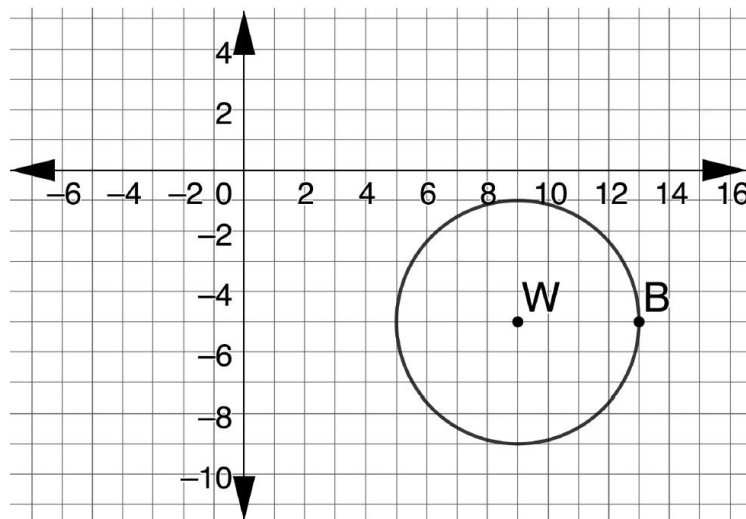
12.7.4 Wrap-Up: Ticket Out the Door

Component Overview: Students apply their knowledge of writing the equation of a circle, given a graph. Students write the key features of a circle, then write the equation of the circle.



12.7.4 Wrap-Up: Ticket Out the Door

1. Circle W is shown, where point B is on the circle.



- a. Determine the key features of circle W .

Key Features	
Radius	4
Center	(9, -5)

- b. Write the equation that represents circle W .

$$(x - 9)^2 + (y + 5)^2 = 16 \text{ or}$$

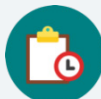
$$(x - 9)^2 + (y + 5)^2 = 4^2$$



Discuss any questions the students still might have regarding this lesson. Consider a Parking Lot approach with a list of the questions on the board or poster in which students can either answer each other's questions to replace a missing assignment grade, or you can revisit as a class. This will give you a quick way to assess how your students are understanding this lesson and its material.

12.8 General Form of a Circle

Lesson Introduction

 Benchmarks	MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features. MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of context.			 Learning Targets	- I can rewrite the equation of a circle from general form to center-radius form. - I can determine key features of a circle given the equation in general form. - I can write the domain and range using inequality notation, interval notation, or set-builder notation.
 Benchmark Coherence	Building On: MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable, quadratic equations over the real number system. Within this benchmark, the expectation is to solve by factoring techniques, taking square roots, the quadratic formula, and completing the square.			Working Toward: N/A	
 Essential Questions	- How can I convert to general form of a circle when it is given in center-radius form? - How do I use completing the square to rewrite the equation of a circle?				
 Lesson Narrative	Students begin to see the equation of a circle in general form. This lesson activates prior knowledge regarding completing the square to show how to convert from general form of a circle to center-radius form of a circle. Students will be able to convert from either form of a circle and identify the key features of a given circle.				
 Component Summary	12.8.1 Warm-Up (~5 min.)	Conic sections discovery (SE p. 267)	12.8.4 Guided Instruction (10 min.)	Conversion from the general form to center-radius form (SE p. 270)	
	12.8.2 Collaboration (10 min.)	Discovering general form (SE p. 267)	12.8.5 Guided Practice (10 min.)	Converting forms of a circle (SE p. 271)	
	12.8.3 Refresher (10-15 min.)	Completing the square (SE p. 268)	12.8.6 Wrap-Up (~5 min.)	Self-Reflection on understanding of key concepts (SE p. 272)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - General Form <u>Additional Terms:</u> - Center-Radius Form		(Optional) Algebra Tiles		12.8.3 Refresher reviews completing the square. Consider preparing your classroom-specific vocabulary or glossary strategies for support (Word Wall, Math Cards, etc.).

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may have forgotten algebraic concepts such as completing the square or factoring perfect square trinomials to convert the circle equation from center-radius form to general form. - Students may misunderstand key or additional terms. - Students may miscalculate when completing the square or factoring.	- Review perfect squares and how to simplify square roots. - Give students several examples of how to graph a circle, given its equation in general form, and how to find the equation of a circle, given its graph. Be sure to give examples of circles whose center coordinates are both negatives, and whose center coordinates are a combination of both positive and negative, as well as examples of circles whose center coordinates are at the origin or are on one of the axes. - Determine if your students need the support of all or part of the 12.8.3 Refresher.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time Consider using this instructional routine throughout this lesson. Give students a few minutes to complete questions individually. Then, allow students time to talk with a partner about their circle and equation. This will allow students to ask clarifying questions, if needed, and have a person to share their ideas with. Students will then fix their equation of a circle, if needed. Consider having students do this throughout the lesson so students can get stronger with writing equations of a circle in general form.	MA.K12.MTR.5.1: Patterns and Structure 12.8.3 Refresher reviews completing the square, which is essential in converting equations of circles in center-radius form to general form. Mathematicians who use patterns and structure to help understand and connect mathematical concepts focus on relevant details within a problem. While completing 12.8.3 Refresher, encourage students to see the patterns and structure to help them understand and connect the completing the square concept. This supports students in developing generalizations based on the similarities found among problems.
 Differentiate Instruction	Consider using additional visuals, such as algebra tiles, to help students better conceptualize the concept completing the square to rewrite the equation of a circle for visual and kinesthetic entry-point. Students will have the chance to draw their own conclusions and discuss their rationale with peers, providing an opportunity for auditory learners to verbalize their understanding. Students can also utilize Study Expert videos from Algebra 1 lessons covering MA.912.AR.3.1 and completing the square for additional scaffolding (in Algebra 1 Unit 8 Lessons 5-7).	
Lesson Notes:		

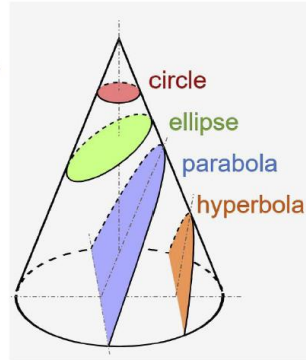
12.8.1 Warm-Up: Conic Sections

Component Overview: Students begin constructing meaningful entry-points into mathematical discussions on the concept of conic sections.



12.8.1 Warm-Up: Conic Sections

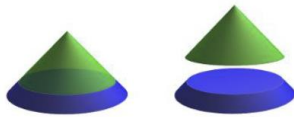
Conic sections are the family of curves created when a plane slices through a cone. A circle is best known as a conic section. A diagram of five conic sections is shown.



According to Greek philosopher Proclus, Menaechmus, a member of Plato's Academy, discovered conic sections around 350 B.C. Approximately 100 years later, Apollonius, a Greek geometer and astronomer, developed and published the theory of conic sections. His work earned him the title of the Great Geometer.



Consider displaying cone, circle, ellipse, parabola, and hyperbola with interactive digital manipulatives, such as GeoGebra.



1. Use the diagram to explain why when a cone is sliced horizontally the result is a circle.
Answers will vary. A horizontal slice is parallel to the base and will have the same shape as the base, which is a circle.

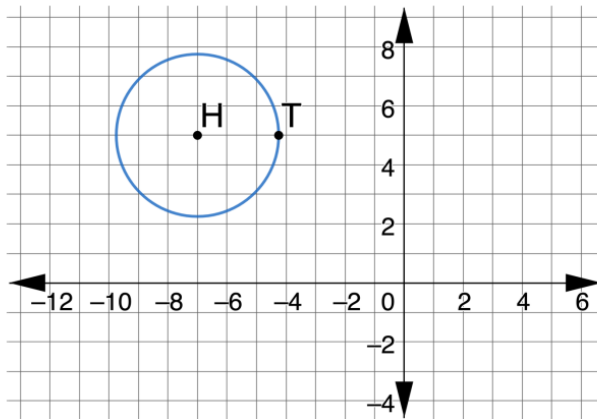
12.8.2 Collaboration: General Form

Component Overview: Students are writing the equation of a circle in center-radius form using the given center and point. Students are then rewriting the equation of the circle in general form.



12.8.2 Collaboration: General Form

- The center of circle H is located at $(-7, 5)$. Point T lies on circle H at the point $(-4.25, 5)$.



- Write the equation of the circle in center-radius form,
 $(x - h)^2 + (y - k)^2 = r^2$.
 $(x + 7)^2 + (y - 5)^2 = 7.5625$ or
 $(x + 7)^2 + (y - 5)^2 = 2.75^2$
- Expand each term of the center-radius form to write an equivalent expression.

Term	Equivalent Expression
$(x + 7)^2$	$x^2 + 14x + 49$
$(y - 5)^2$	$y^2 - 10y + 25$
$(2.75)^2$	7.5625



The **general form** of an equation of a circle is
 $ax^2 + by^2 + cx + dy + e = 0$.

- Use the equivalent expressions of the center-radius form to rewrite the center-radius form of circle H in general form.
 $x^2 + y^2 + 14x - 10y + 66.4375 = 0$



As students collaborate, listen for opportunities to prompt mathematical discussion. Elicit any exemplar student responses whole-group after the component.

- If a coordinate in the center is positive, will the equation always, sometimes, or never have a subtraction sign in front of the positive value? Explain.
- How can you find the radius of the circle?
- Can you think of another way to find the radius of the circle? (Distance Formula or plugging in point T into the equation.)



Ask your students to explain what information is needed in order to write the equation of a circle. Have them explain how you could verify your equation of a circle is correct.

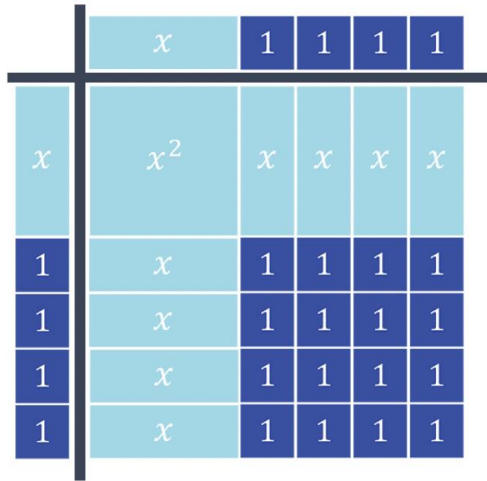
12.8.3 Refresher: Completing the Square

Component Overview: Students review completing the square using a pictorial representation of algebra tiles. Students are asked to complete the square by finding the perfect square trinomial. This component will allow students to be able to go from general form to center-radius form.



12.8.3 Refresher: Completing the Square

The binomial $(x + 4)^2$ can be rewritten as $x^2 + 8x + 16$, which is a perfect square trinomial. The algebra tiles represent both $(x + 4)^2$ and $x^2 + 8x + 16$.



1. Discuss with your partner how the diagram shows that $x^2 + 8x + 16$ is a perfect square trinomial. Summarize your discussion.

Answers may vary.

The shape of the algebra tiles make a square.

2. Discuss with your partner how the diagram shows that $x^2 + 8x + 16 = (x + 4)^2$.

The outsides of the algebra tiles show that $(x + 4)(x + 4) = (x + 4)^2$.



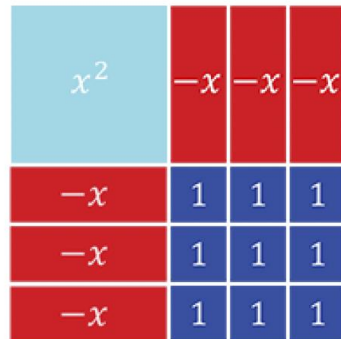
Students must be able to complete the square to effectively engage in 12.8.4 Guided Instruction. 12.8.3 Refresher provides embedded scaffolding for students to review their working knowledge of MA.912.AR.3.1 and completing the square.



Consider splitting the class into groups for this component and pulling a small group for students that will struggle with this content.

12.8.3 Refresher: Completing the Square (continued)

3. Write the perfect square trinomial and the binomial squared that are represented by the diagram.

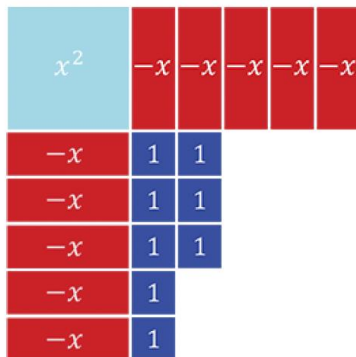


$$x^2 - 6x + 9 = (x - 3)^2$$



Provide hands-on or digital manipulatives for students to engage with algebra tiles for kinesthetic and visual entry-points.

4. The trinomial $x^2 - 10x + 8$ is shown.



- a. How many 1 tiles have to be added to complete the square?

17

12.8.3 Refresher: Completing the Square (continued)

Completing the square by adding 1 tiles will change the value of the original trinomial unless the same number of -1 tiles are also added.

b. Complete the table.

Completing the Square	
Original Expression	$x^2 - 10x + 8$
Completing the square	$x^2 - 10x + 8 + 17$ number of 1 tiles added
Adding the opposite so the expression is equivalent to the original expression	$x^2 - 10x + 8 + 17 + -17$ number of -1 tiles added
Rewriting so that there is a perfect square trinomial	$(x^2 - 10x + 25) - 17$ perfect square trinomial
Writing the perfect square trinomial as a binomial squared	$(x - 5)^2 - 17$



Make a meaningful connection between 12.8.3 Refresher and 12.8.4 Guided Instruction. Why are we reviewing something that we learned in Algebra 1? The learning experience of this lesson should remain centered on the concept of converting between center-radius form and general form.

Students who are able to articulate why completing the square matters in the equations of circles may struggle with the algebraic operation of completing the square, but should be recognized for their learning from the lesson.

12.8.4 Guided Instruction: Rewriting the General Form of the Equation of a Circle

Component Overview: Students are completing the square to rewrite the equation of a circle from general form to center-radius form. Students are identifying the center and radius of each equation of a circle.



12.8.4 Guided Instruction: Rewriting the General Form of the Equation of a Circle

There are several ways to mathematically represent the domain and range.

Written Description	Inequalities	Set-Builder Notation	Interval Notation
All real numbers greater than or equal to -5 and less than or equal to 5	$-5 \leq x \leq 5$	$\{x: -5 \leq x \leq 5\},$ $\{x -5 \leq x \leq 5\},$ $\{x: x \geq -5 \text{ and } x \leq 5\},$ or $\{x x \geq -5 \text{ and } x \leq 5\}$	$[-5, 5]$

Rewrite each general form equation of a circle to center-radius form of a circle. Then, give the key features of the circle. Write the domain and range using interval notation and set-builder notation.

1. $x^2 + y^2 - 14x + 6y + 26 = 0$
 $(x^2 - 14x + 49) + (y^2 + 6y + 9) = -26 + 49 + 9$
 $(x - 7)^2 + (y + 3)^2 = 32$

Center-Radius Form	Center	Radius
$(x - 7)^2 + (y + 3)^2 = 32$	$(7, -3)$	$\sqrt{32}$ or $4\sqrt{2}$
	Domain	Range
	$\{x: (7 - \sqrt{32}) \leq x \leq (7 + \sqrt{32})\}$ $[(7 - \sqrt{32}), (7 + \sqrt{32})]$	$\{y: (-3 - \sqrt{32}) \leq y \leq (-3 + \sqrt{32})\}$ $[(-3 - \sqrt{32}), (-3 + \sqrt{32})]$



Provide algebra tiles for additional entry-points when completing the square.



Make connections between 12.8.3 Refresher and 12.8.4 Guided Instruction so these activities do not feel isolated:

- How does this equation look the same as what we just did in 12.8.3 Refresher?
- How does it look different?
- Why might we need to complete the square here?



You may need to refresh students on how to write square roots in simplest form for the radius.

12.8.4 Guided Instruction: Rewriting the General Form of the Equation of a Circle (continued)

2. $x^2 + y^2 - 18x - 10y = -52$
 $(x^2 - 18x + 81) + (y^2 - 10y + 25) = -52 + 81 + 25$
 $(x - 9)^2 + (y - 5)^2 = 54$

Center-Radius Form	Center	Diameter
$(x - 9)^2 + (y - 5)^2 = 54$	$(9, 5)$	$2(\sqrt{54})$ or $6\sqrt{6}$
	Domain	Range
	$\{x: (9 - \sqrt{54}) \leq x \leq (9 + \sqrt{54})\}$ $[(9 - \sqrt{54}), (9 + \sqrt{54})]$	$\{y: (5 - \sqrt{54}) \leq y \leq (5 + \sqrt{54})\}$ $[(5 - \sqrt{54}), (5 + \sqrt{54})]$

12.8.5 Guided Practice: Converting Forms of a Circle

Component Overview: Students begin to formalize the process of converting forms of a circle. This component gives students the opportunity to apply the conversion to both center-radius form and general form.



12.8.5 Guided Practice: Converting Forms of a Circle

1. Consider the equation of the circle represented by $(x + 8)^2 + (y + 11)^2 = 14^2$.

a. Rewrite $(x + 8)^2 + (y + 11)^2 = 14^2$ in general form.
 $x^2 + y^2 + 16x + 22y - 11 = 0$

- b. Give the key features of $(x + 8)^2 + (y + 11)^2 = 14^2$

Center	Diameter	Radius
$(-8, -11)$	28	14
Domain	Range	
$-22 \leq x \leq 6$	$-25 \leq y \leq 3$	

2. Consider the equation of the circle represented by $x^2 + y^2 - 20x = -36$.

a. Rewrite $x^2 + y^2 - 20x = -36$ in center-radius form.
 $(x^2 - 20x + 100) + y^2 = -36 + 100$
 $(x - 10)^2 + y^2 = 64$

- b. Give the key features of $x^2 + y^2 - 20x = -36$

Center	Diameter	Radius
$(10, 0)$	16	8
Domain	Range	
$[2, 18]$	$[-8, 8]$	



Think-Pair-Share

Consider conducting the Think-Pair-Share instructional routine for 12.8.5 Guided Practice. Students will make better sense of thinking and reasoning with the opportunity to independently process then refine answers with a partner, before the teacher elicits more precise responses as a whole group.



Use inquiry as students work to synthesize and apply their learnings from 12.8.3 Refresher and 12.8.4 Guided Instruction:

- What form of the circle is given? How do you know?
- How can you convert from center-radius form to general form?
- How can you identify the center, diameter, and radius of a circle?



Clarify, Critique, Correct

Consider the Clarify, Critique, and Correct instructional routine as a way to deepen your students' understanding. Have the students exchange their work for these two problems to analyze, reflect, and develop their partner's work. This will prompt student reflection, whether it is for students to improve upon, correct errors, or clarify their partner's work.

12.8.6 Wrap-Up: Reflection

Component Overview: Students complete a self-reflection on their level of understanding of the content of this lesson.



12.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

Rewriting center-radius form to general form			
Key features of a circle			
Completing the square			
Rewriting general form to center-radius form			
Set-builder notation			
Interval notation			

Answers may vary.



Note that completing the square is only one of four targets within 12.8.6 Wrap-Up. During the lesson, continue centering students' learning experiences on the concept of changing from center-radius to general form, rather than getting stuck on the algebraic procedure of completing the square.

12.9 Graphing Circles

Lesson Introduction

 Benchmark	MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of context.		 Learning Targets	- I can graph a circle and determine its key features when given an equation of a circle. - I can solve mathematical and real-world problems that are modeled with an equation of a circle.	
 Benchmark Coherence	Building On: N/A			Working Toward: N/A	
 Essential Questions	- What are the two key elements of a circle needed in order to graph the circle? - How can you graph a circle when the equation is written in general form?				
 Lesson Narrative	Throughout this lesson, students will be using key features of a circle to graph a circle on the coordinate plane. This lesson activates prior knowledge regarding domain and range, and requires students to write inequalities based on the equation of the circle. Students are given real-life scenarios and asked to answer questions based on the information given in the problems.				
 Component Summary	12.9.1 Warm-Up (~5 min.)	Notice and Wonder with U.S. Capitol Rotunda (SE p. 273)	12.9.3 Your Turn! (15 min.)	Independent practice graphing circles (SE p. 275)	
	12.9.2 Guided Practice (25 min.)	Graphing circles practice (SE p. 273)	12.9.4 Wrap-Up (~5 min.)	Error Analysis graphing circles (SE p. 277)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Domain - Range - Center - Radius		N/A		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may struggle with relating the key features of a circle to the context of the question. For example, students may struggle making connections in question 2 during 12.9.2 Guided Practice.	Students may have forgotten algebraic concepts such as how to find the domain, range, and writing inequalities. Consider small group instruction and differentiating routines as a primary driver of learning in this lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time 12.9.2 Guided Practice provides a structured and interactive opportunity for students to revise and refine both their ideas and their verbal and written output. The main idea of this routine is to have students think or write a response individually, and then use structured pairings for iterations of feedback and revisions, providing multiple opportunities to refine and clarify the response.	MA.K12.MTR.7.1: Real-World Contexts This lesson provides the opportunity for students to connect mathematical concepts to everyday experiences. Consider encouraging students to apply mathematics to real-world contexts, and support students as they validate conclusions by comparing them to the given situation.
 Differentiate Instruction	This lesson requires the student to be able to activate prior knowledge of domain, range, and writing inequalities. Consider reviewing these critical prerequisite concepts through both demonstration and models. For graphing and writing inequalities, there are two On-Ramp videos that can act as scaffolding to be used before or during the lesson. Even though this lesson focuses on the not-strict inequalities, having the students review the key differences of both will be helpful.	
Lesson Notes:		

12.9.1 Warm-Up: Notice and Wonder

Component Overview: Students will make observations and write down things they notice and wonder about the dome of the U.S. Capitol Building.



12.9.1 Warm-Up: Notice and Wonder

The dome of the U.S. Capitol Building is 288 feet (ft) in height and 96 ft in diameter. The dome covers the Rotunda, a room with a circular plan. The architect created an orchestration of architecture, sculpture, and painting in the U.S. Capitol Rotunda.



This component provides the opportunity for interdisciplinary studies. Consider challenging students by having them incorporate information about the U.S. Capital Building to lessons they have studied in their Civics or other Social Studies classes.

1. What do you notice?

Answers may vary.

I notice that the rotunda has concentric circles.

2. What do you wonder?

Answers may vary.

I wonder what is the diameter of the innermost circle.

12.9.2 Guided Practice: Graphing Circles

Component Overview: Students are working with the equation of a circle in general form, and are asked to identify the key features such as center, radius, domain, and range. This will lead students into making a connection between key features of a circle and the context of a real-world problem. Consider giving students space and opportunities for analyzing errors, engaging in math talks, and collaborating in order to orient students to the content and each other.



12.9.2 Guided Practice: Graphing Circles

1. The equation of circle W is shown.

$$x^2 + y^2 - 4x + 10y + 13 = 0$$

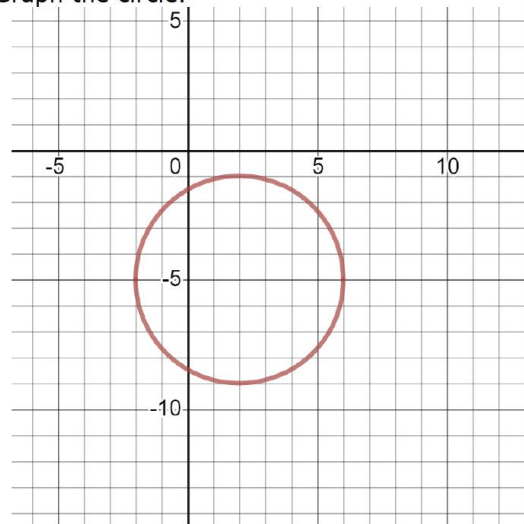
- a. Write the equation in center-radius form.

$$(x - 2)^2 + (y + 5)^2 = 16$$

- b. Complete the table.

Center	Radius
$(2, -5)$	4

- c. Graph the circle.



- d. Find the domain of $x^2 + y^2 - 4x + 10y + 13 = 0$. Write it as an inequality.

$$-2 \leq x \leq 6$$

- e. Find the range of $x^2 + y^2 - 4x + 10y + 13 = 0$. Write it as an inequality.

$$-9 \leq y \leq -1$$



Number Talk

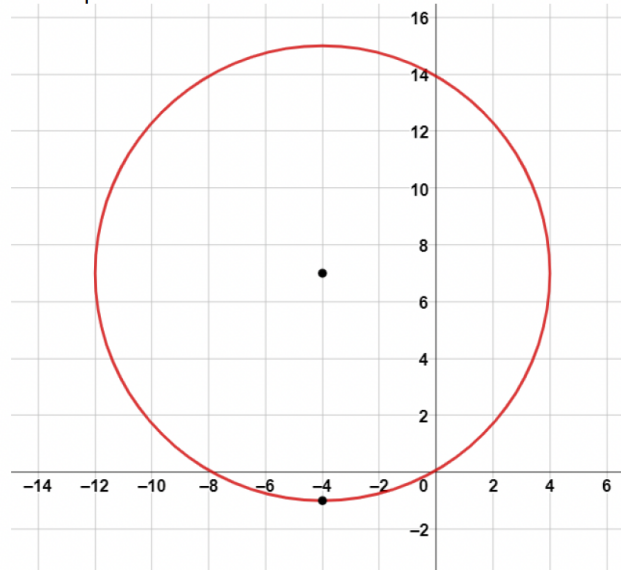
Display one question at a time. Give students think time before prompting a signal when they have an answer or strategy on solving each part. Students should find a partner based on shared signals and explain their findings.

Then, the teacher selects students to share different strategies for each part. ("Who thought about it in a different way?") Their explanations are recorded for all to see. Students might be pressed to provide more details about why they decided to approach a part of the question a certain way.

12.9.2 Guided Practice: Graphing Circles (continued)

2. Marcus wanted to order from a food delivery service and used the equation $(x + 4)^2 + (y - 7)^2 = 64$ of circle M to find food within his radius, in miles.

a. Graph circle M .



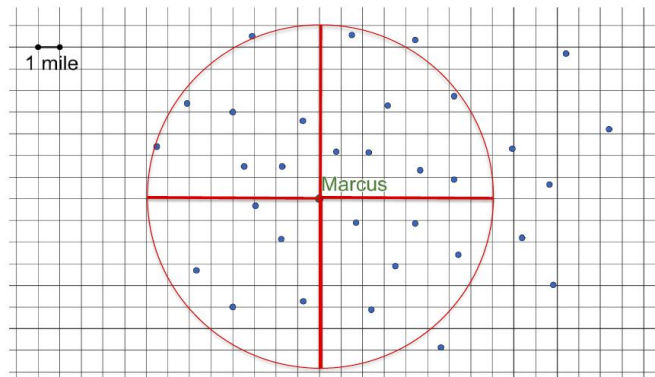
Monitor as students work through each question. Use inquiry to extend or scaffold student thinking when appropriate:

- What does the domain of a circle mean?
- What does the range of a circle mean?
- Why is the domain and range needed for Marcus' delivery?

- b. What does the center represent in the context?
The center represents the location of Marcus's house.
- c. What does the radius represent in the context?
The radius represents the distance in miles and area from which Marcus can order food.
- d. If the service charges \$3.25 for delivery for every four miles driven, what is the most that Marcus could be charged for the delivery?
\$6.50

12.9.2 Guided Practice: Graphing Circles (continued)

- e. Determine the domain of $(x + 4)^2 + (y - 7)^2 = 64$. Write the domain in set-builder notation.
 $\{x | -12 \leq x \leq 4\}$
- f. Determine the range of $(x + 4)^2 + (y - 7)^2 = 64$. Write the range in set-builder notation.
 $\{y | -1 \leq y \leq 15\}$
- g. The grid shows Marcus' location in green. The blue dots are the locations of restaurants that deliver using UberEats. Use the grid to demonstrate how Marcus could use the domain and range to determine which restaurants he can order from.



The domain and range will tell Marcus which restaurants he is able to place an order within his area.



Use higher-order thinking questions to informally measure concept mastery. These questions can be used as an alternative oral assessment for specific students. The following questions could be used:

- In your own words, explain the steps in converting an equation of a circle in center-radius form to general form. Create/produce an example to use in your explanation.*
- Explain how you could write the equation of a circle in center-radius and general forms if you were given only the endpoints of the diameter. Create/produce an example to use in your explanation.*
- Create/produce three truths and a lie about converting an equation of a circle from general form to center-radius form.*



What radius will Marcus need to be able to order from all the restaurants shown on the coordinate plane?

Why do you think food delivery services have a limit to where they deliver to?

12.9.3 Your Turn!

Component Overview: Students are given the opportunity to take what they learned in 12.9.2 Guided Practice and apply it independently. Probe students as they work, using assessing and critical thinking questions to check for understanding, to challenge students, and to help students who are struggling. Consider a peer review process for students to check each other's work as an added layer of support.



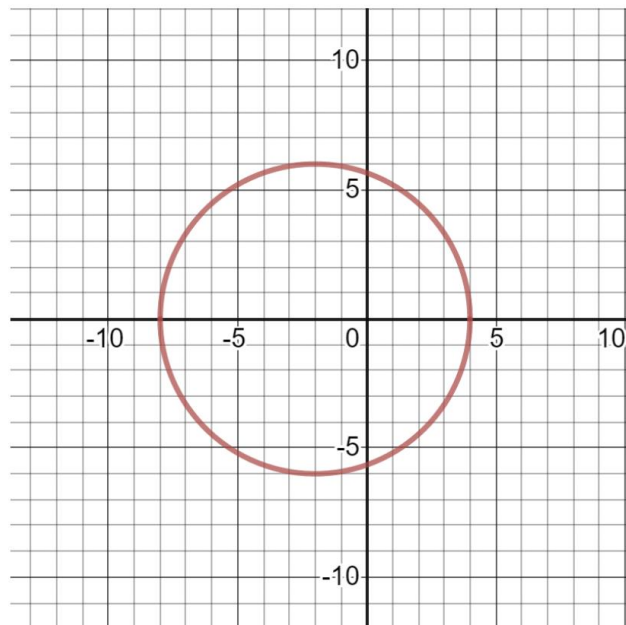
12.9.3 Your Turn!

1. The equation of circle R is $(x + 2)^2 + y^2 = 36$.

a. Complete the table.

Center	$(-2, 0)$
Radius	6
Domain	$-8 \leq x \leq 4$
Range	$-6 \leq y \leq 6$

b. Graph circle R .



Consider prompting questions for students who are unsure how to begin:

- How would you determine what information to use to start this question?
- Do you need to rewrite the equation before you can start? Why or why not?
- Do you need all the information provided? Why or why not?

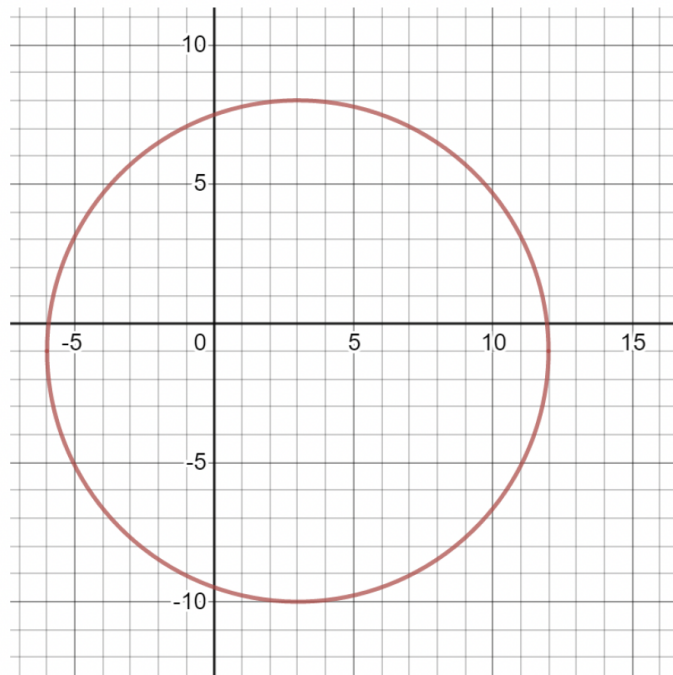
12.9.3 Your Turn! (continued)

2. Alyona used the equation $(x - 3)^2 + (y + 1) = 81$ to determine how far she could sit, in meters, from her Xbox and still use her wireless controller.

a. Complete the table.

Center	$(3, -1)$
Radius	9
Domain	$-6 \leq x \leq 12$
Range	$-10 \leq y \leq 8$

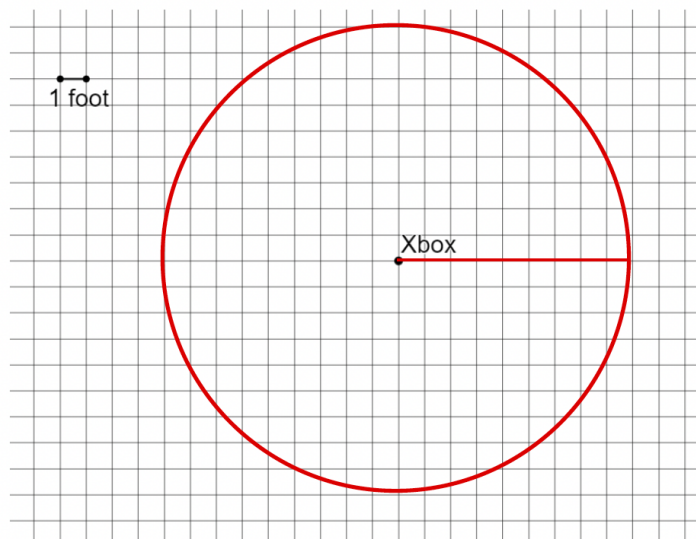
b. Graph $(x - 3)^2 + (y + 1) = 81$



For students who struggled during 12.9.2 Guided Practice, consider using this time to work with individual or small groups of students on interpreting an equation of a circle in real-world situations. Some students might not remember how to identify key features of a circle, and some might not be able to connect them to the real-world situation, so utilizing pairs or small groups will contribute toward a collaborative learning environment.

12.9.3 Your Turn! (continued)

- c. What does the center represent in the context?
The location of the Xbox
- d. What does the radius represent in the context?
How far Alyona can be from the Xbox and still use the wireless controller.
- e. The grid shows the location of Alyona's Xbox. Use the domain and range to show all of the locations at which Alyona can sit.



Consider challenging students who finish early to find the area Alyona's Xbox reaches.

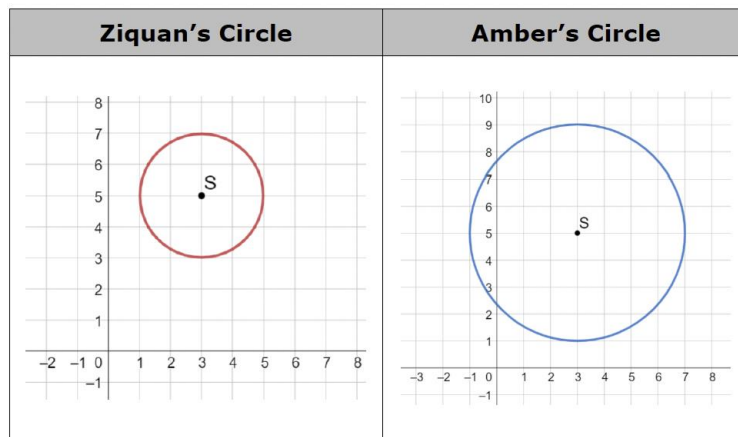
12.9.4 Wrap-Up: Error Analysis

Component Overview: 12.9.4 Wrap-Up assesses students' understanding of the key parts of a circle and how to graph a circle on a coordinate plane. Consider using all parts of this exercise as a formative assessment to gather data on students' progress from this lesson.



12.9.4 Wrap-Up: Error Analysis

1. Ziquan and Amber both graphed circle S , $(x - 3)^2 + (y - 5)^2 = 4$, but their circles looked different.



Visit the activities and Activity Builder on Desmos for a variety of engaging extension ideas for students to engage with following this lesson and unit.

- a. Who correctly graphed circle S ?

Ziquan

- b. Write a note to the person who graphed circle S incorrectly explaining their mistake.

Amber, the equation $(x - h)^2 + (y - k)^2 = r^2$ is the center-radius form of the equation of a circle. In the equation $(x - 3)^2 + (y - 5)^2 = 4$, $r^2 = 4$ so the radius is 2. You graphed a circle with a radius of 4.